



Deutsche Bank
Research

Investors see megatrends as an opportunity

May 2026

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Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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We asked investors to explain why they are positive about investing in megatrends.

Our AI analysis of hundreds of responses is visualised in the word cloud below



This chartbook follows our [report](#) explaining how our new model analysing the impact of megatrends on economies and markets



We thank dbDataInsights for their contribution to this report

Source: dbDataInsights, Deutsche Bank.

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 - Technology
 - Sovereign deficits
 - Geopolitics & globalisation
 - Domestic politics & social discontent
 - Demography
 - Energy disruption & transition

Our survey polled around 500 people in both the US and UK. The sample was designed to be representative of the population – not just a selection of investment professionals.

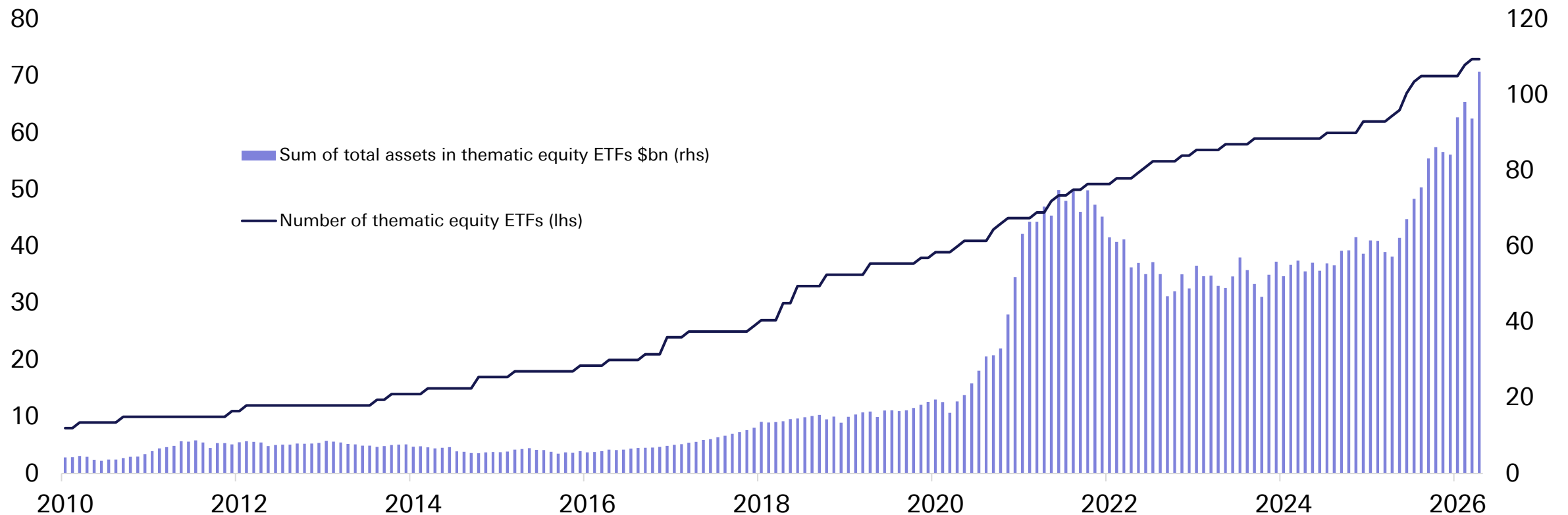
Thematic investing has been particularly strong in the 2020s

AI has also catalysed a rush of money flowing into thematic funds



The rise of thematic investing

Thematic equity funds and their AuM



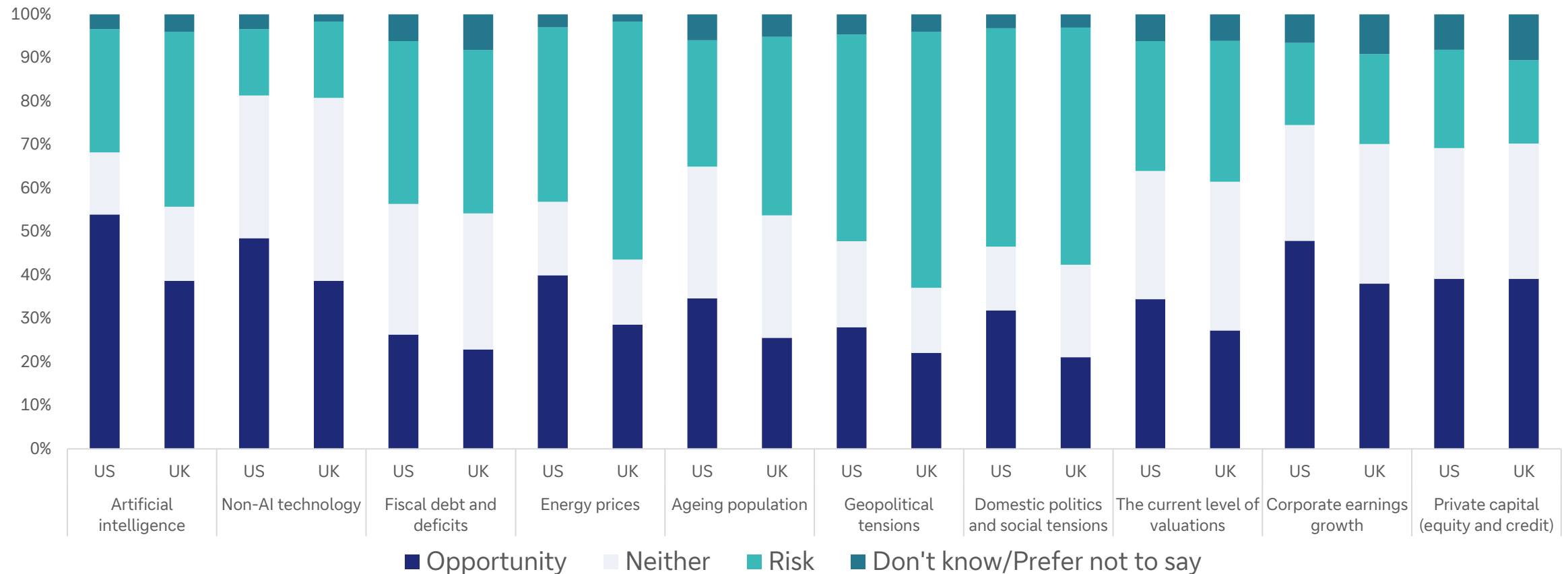
Source: Bloomberg Finance L.P., Deutsche Bank.

Tech – both AI and non-AI – are the strongest themes for investors

The similarity in the popularity of both AI and corporate earnings growth suggests that investors are happy to take a long-term position on AI regardless of the current capex requirements



“For each investment theme listed below, please indicate whether you perceive it as a risk or an opportunity for investors in 2026.”



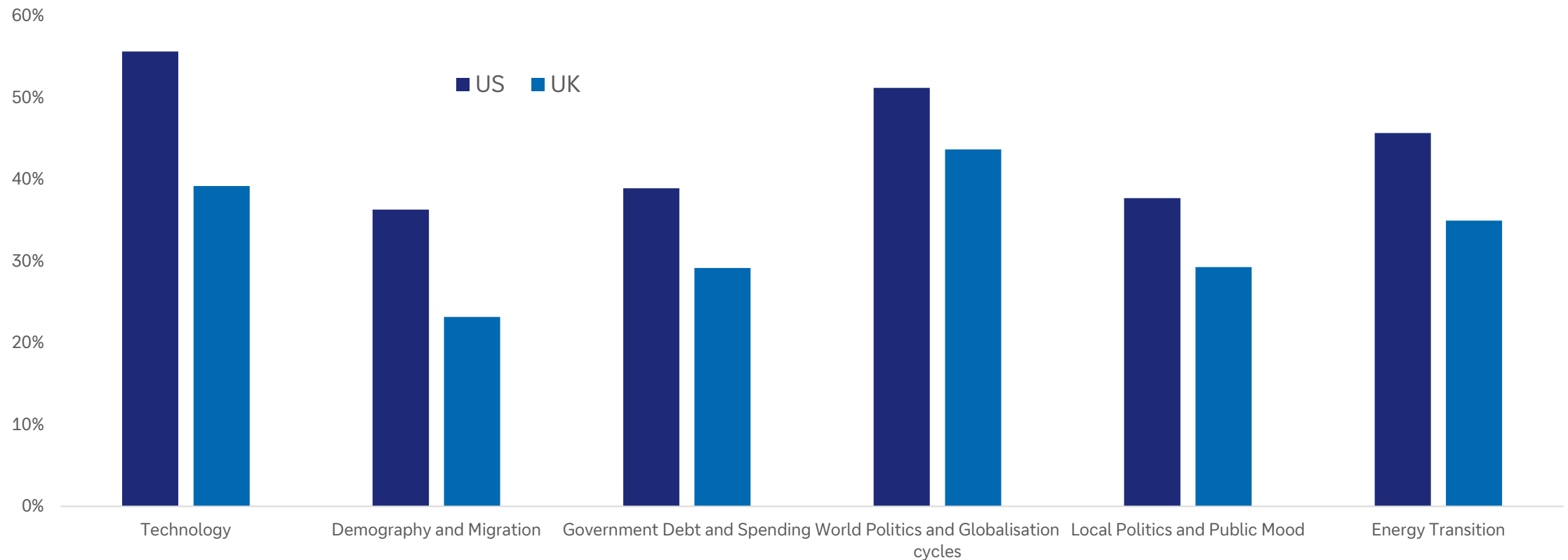
Source: dbDataInsights, Deutsche Bank.

Active management: Investor attention to megatrends seems to contradict the narrative that everyone is piling into passive indices.

Americans are more likely to actively manage their portfolios for megatrends, particularly for technology



Proportion of investors who have made a change to their investment portfolio in the last three years as a result of the following megatrends



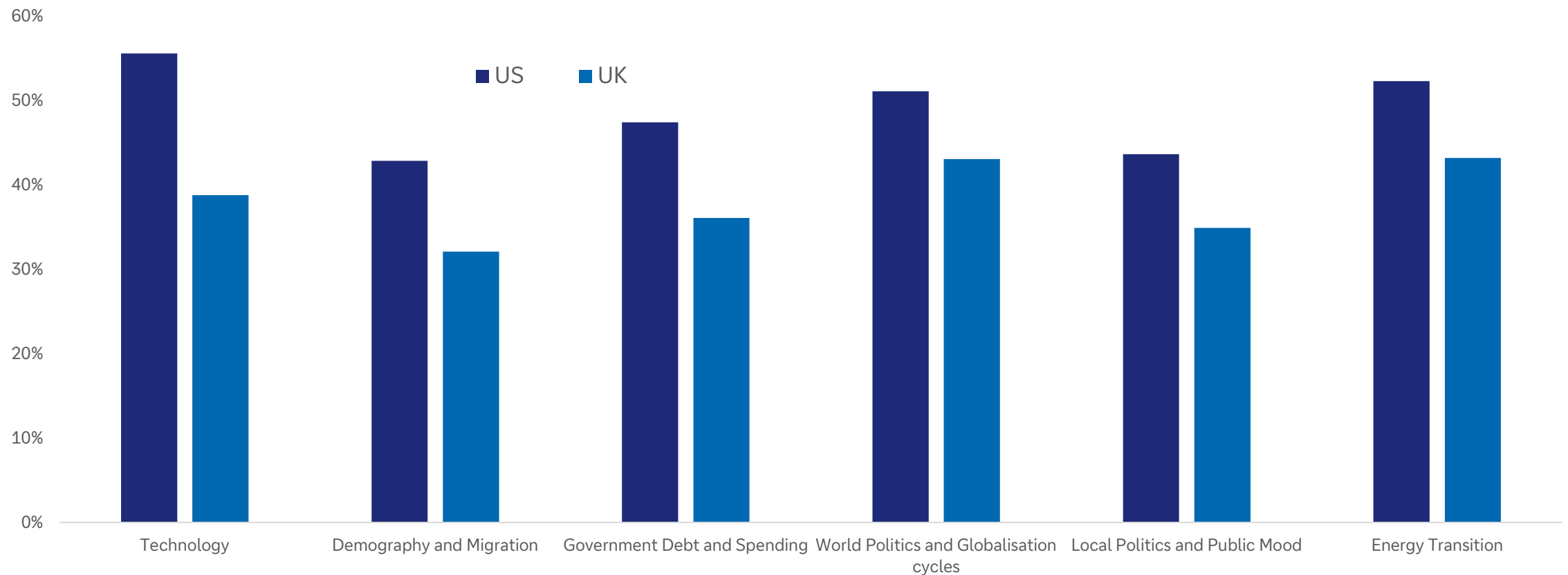
Source: dbDataInsights, Deutsche Bank.

Active management: The volatility of the last two years perhaps a key reason why investors continue to expect to make investment changes over the coming 12 months



No doubt volatility of global politics feeds into the decisions; however, the pace of AI adoption is also key

Proportion of investors who are 'likely' or 'very likely' to change their investment portfolio in the next 12 months as a result of the following megatrends



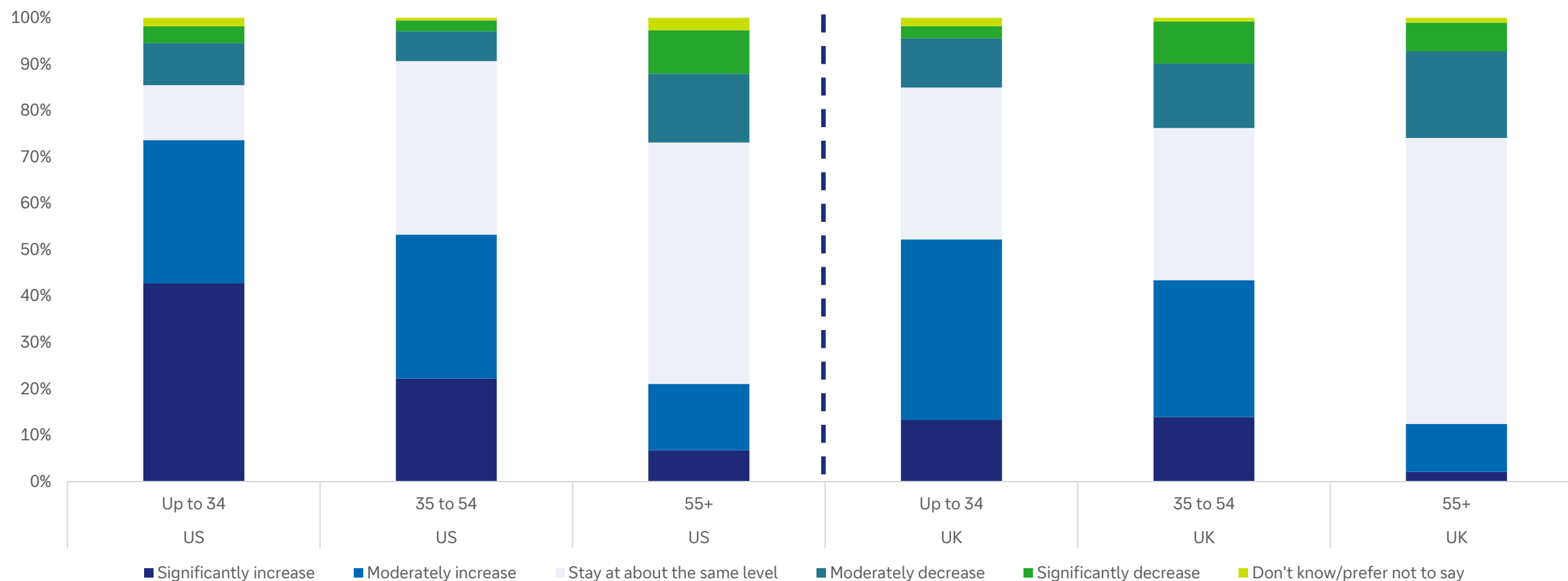
Source: dbDataInsights, Deutsche Bank.

Risk is back: Far more people expect to *increase* their risk taking over the coming 12 months

Despite the four major international shocks that have affected markets this decade, investors appear undeterred. Partly, this is due to enthusiasm about AI, but it may also be partly due to consistent way in which equity markets have rebounded after sharp shocks.



Over the next 1 year, do you expect your level of risk-taking in relation to your main investment portfolio to...

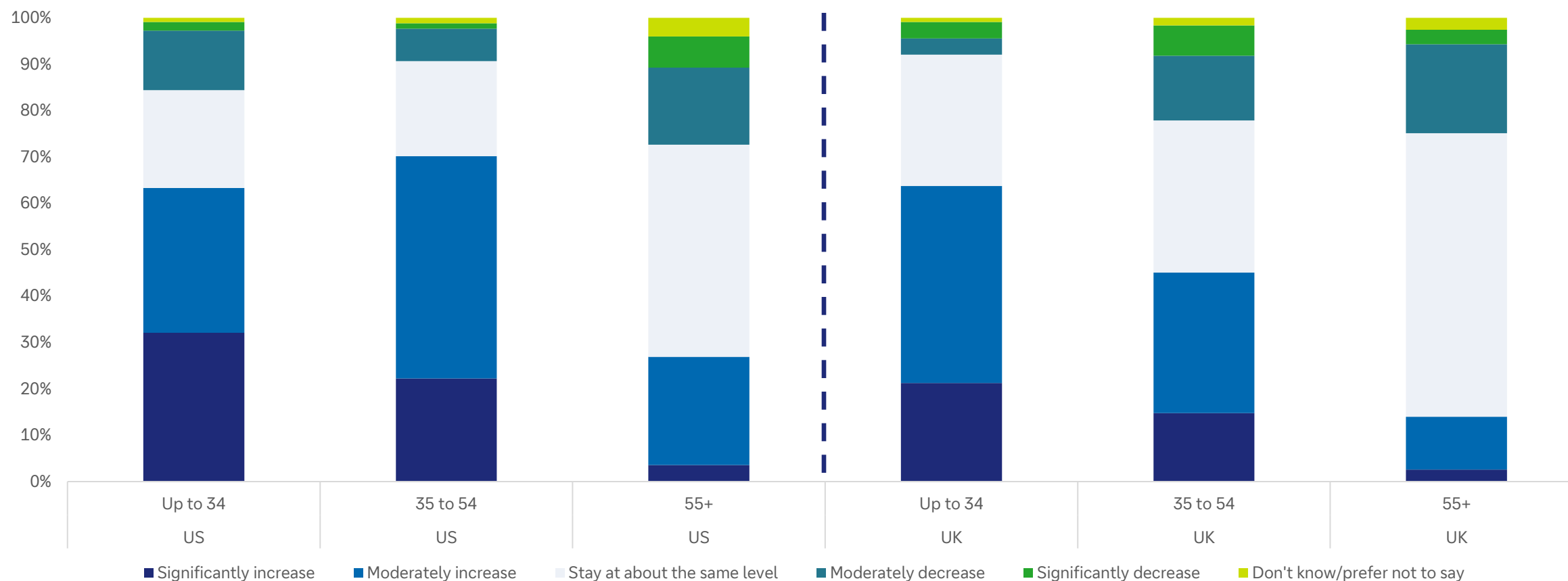


Source: dbDataInsights, Deutsche Bank.

Risk on: Three years out, expectations for risk taking are even stronger ...



Over the next 3 years, do you expect your level of risk-taking in relation to your main investment portfolio to...

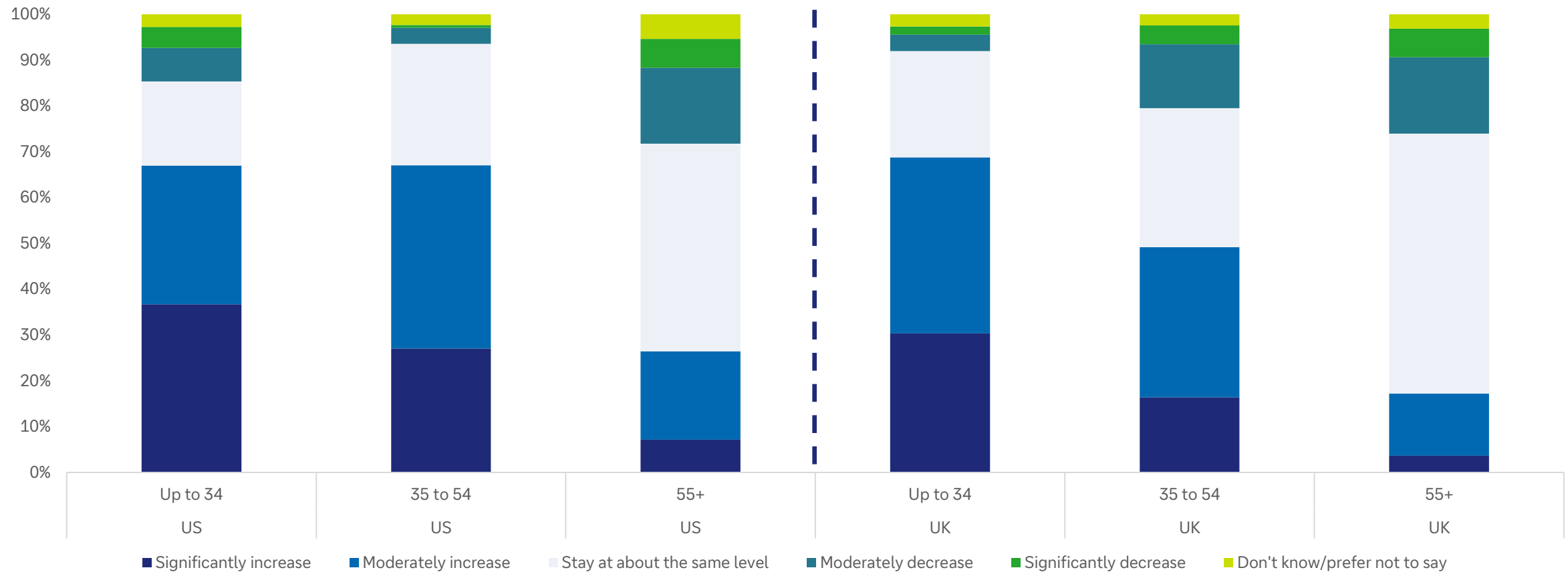


Source: dbDataInsights, Deutsche Bank.

... before risk tolerance plateaus into a 5-year timeframe



Over the next 5 years, do you expect your level of risk-taking in relation to your main investment portfolio to...



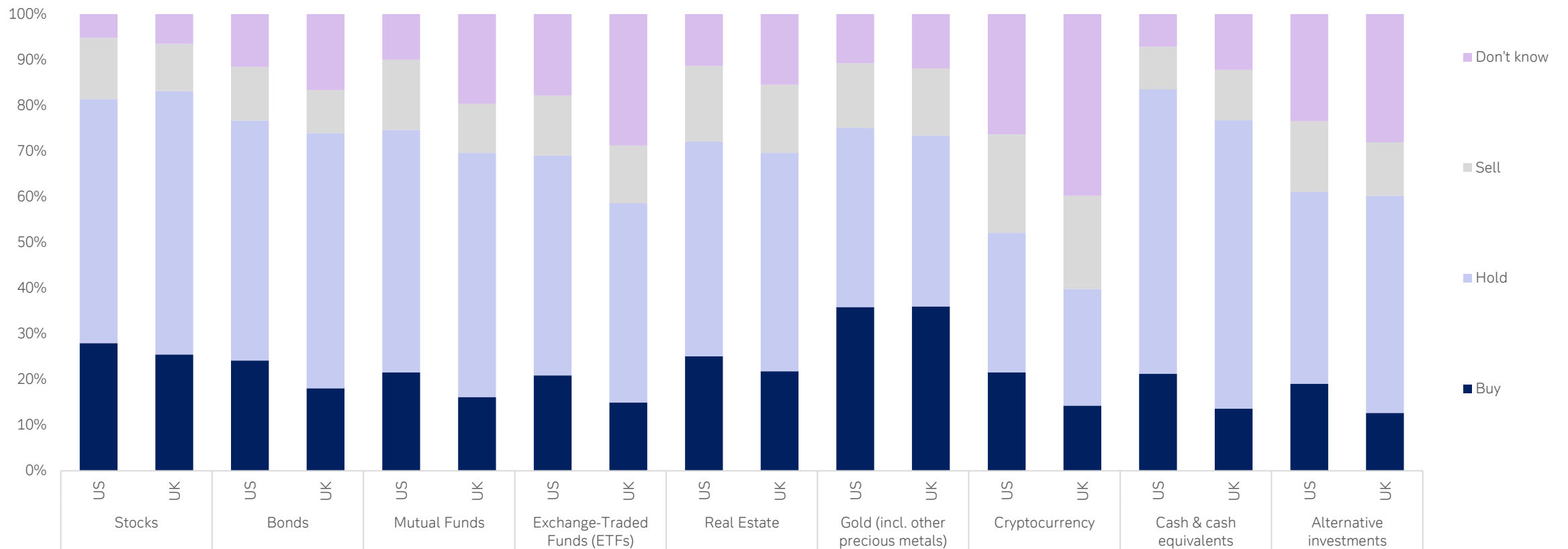
Source: dbDataInsights, Deutsche Bank.

What investors do when things are uncertain

The market volatility of the 2020s has been a great litmus test for how investors react to uncertainty. It is unexpected to see that equities are seen as a stronger 'buy' option than bonds



“In any period of economic uncertainty, what do you think is the best strategy for each of the following investment asset categories?”



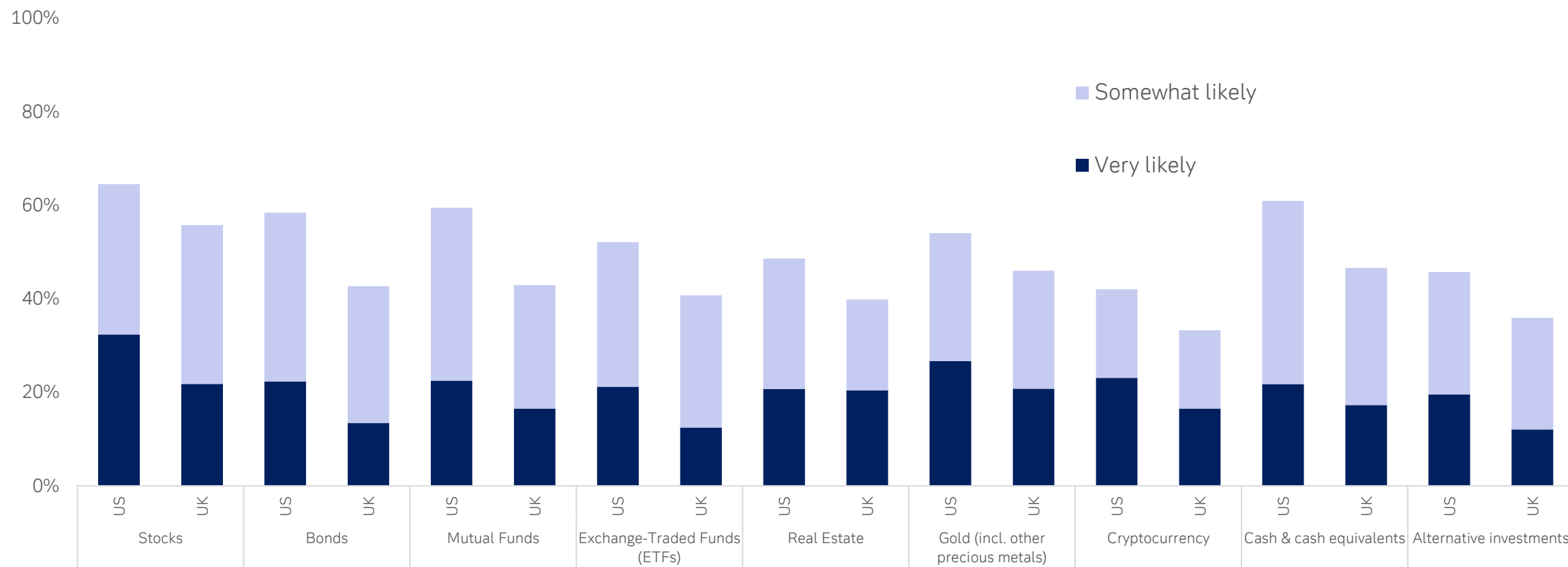
Source: dbDataInsights, Deutsche Bank.

Domestic politics can be highly correlated with economic growth ...

... but the response from individual investors may be more muted than many expect



“If your country’s government were to change in the near future, how likely or unlikely would you be to invest in each of the following asset categories?”



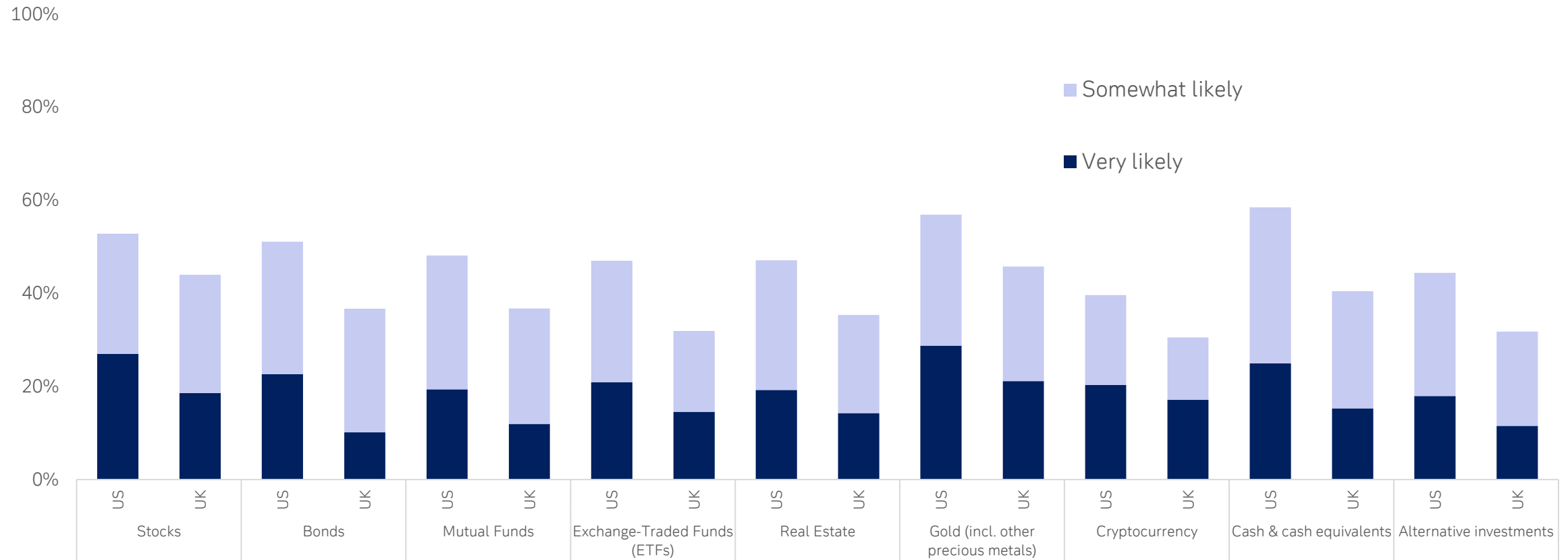
Source: dbDataInsights, Deutsche Bank.

Reacting to geopolitical uncertainty

Gold and cash are no surprise, but it is interesting to see that so many investors see equities as a geopolitical hedge. That may be another reaction to the rapid bounce-backs in equity markets this decade



“If tensions escalated between the US government and its major trading partners, how likely or unlikely would you be to invest in each of the following asset categories?”



Source: dbDataInsights, Deutsche Bank.



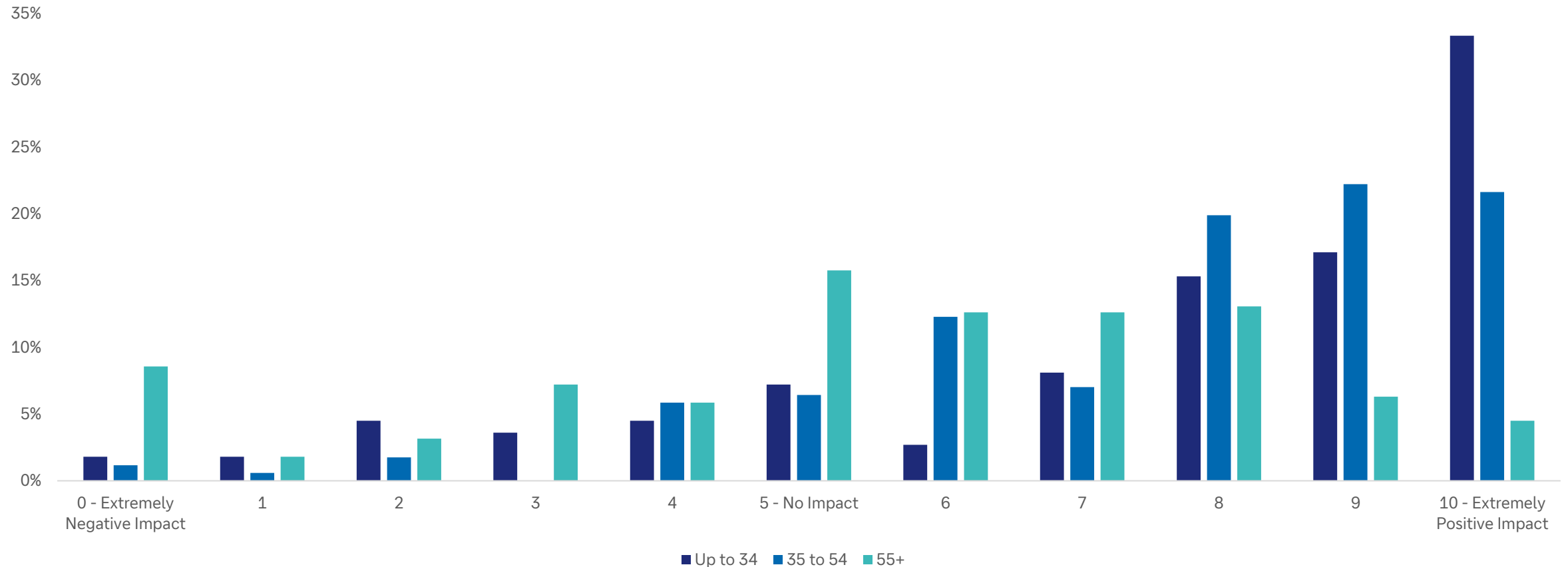
What investors think about individual
megatrends

Technology

Overwhelming optimism is the theme. The question is whether the returns will come from the specific assets to which investors are exposed, or from other more nuanced and diversified sources.



Investors' anticipated impact of technology on their main investment portfolio over the next 3 years (0-10)



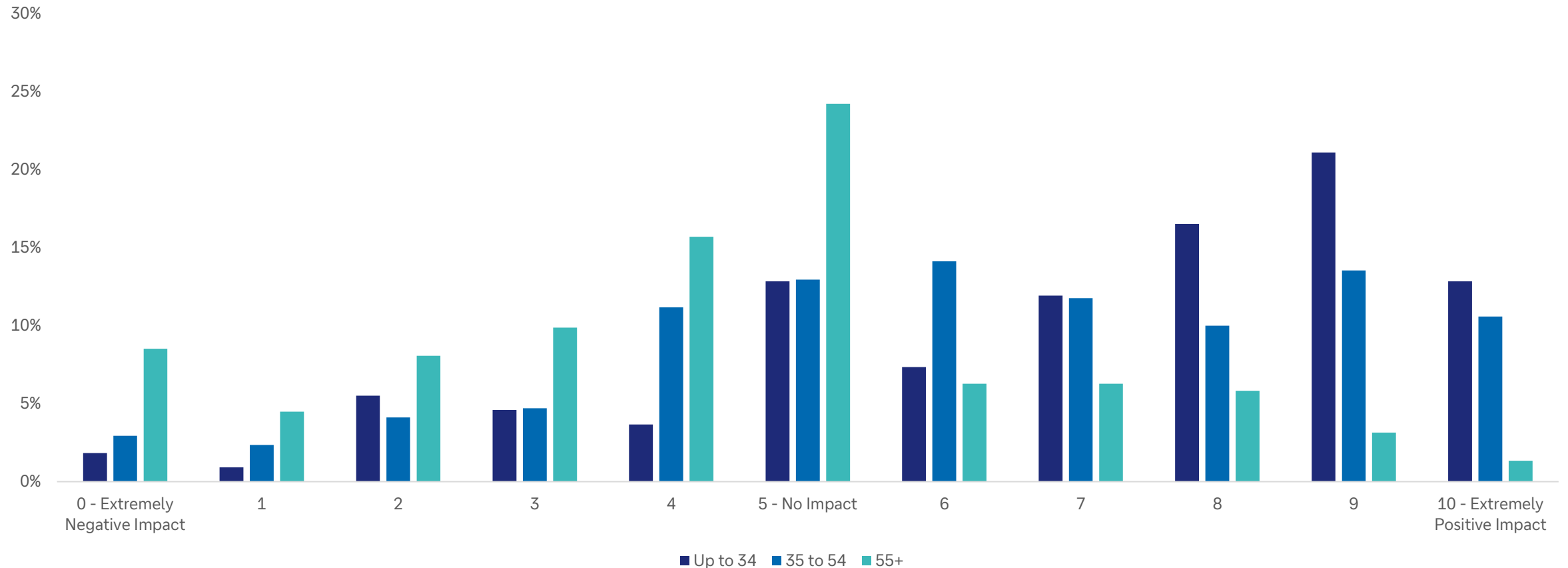
Source: dbDataInsights, Deutsche Bank.

Sovereign deficits

People are more optimistic than we expected about the impact of sovereign deficits. This may reflect an opinion that, somehow, governments will find a way to muddle through their current problems.



Investors' anticipated impact of sovereign deficits on their main investment portfolio over the next 3 years (0-10)



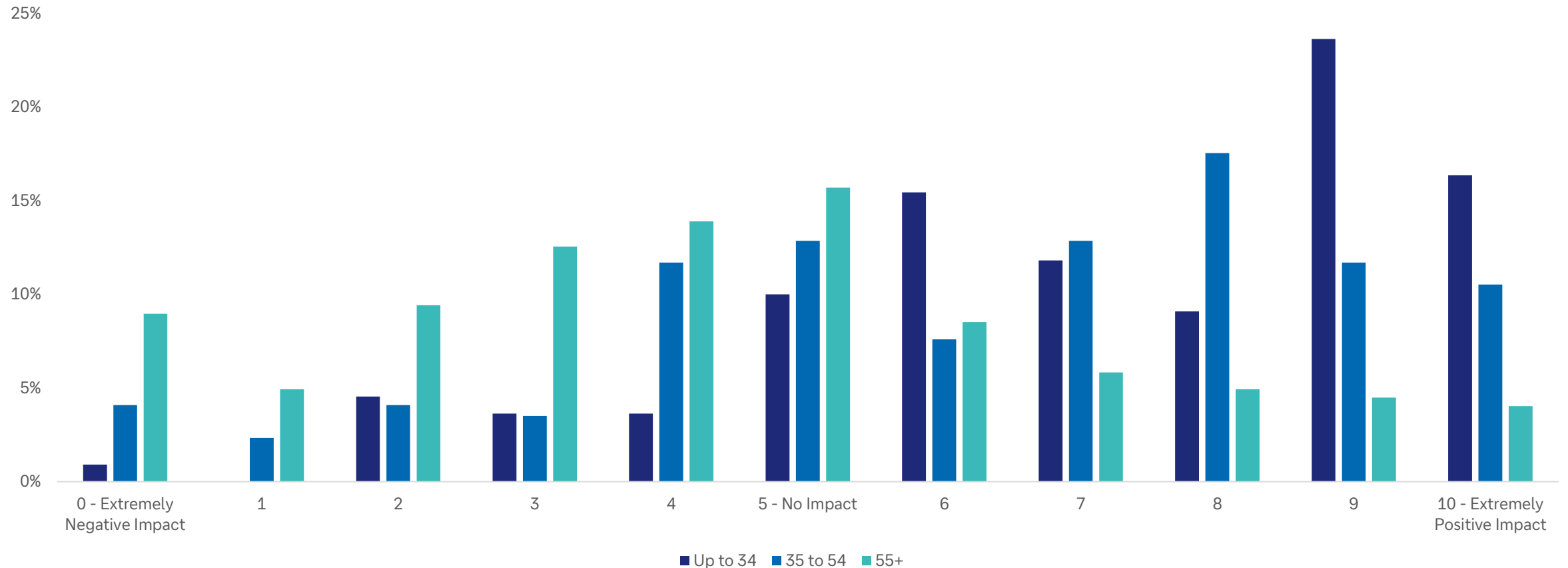
Source: dbDataInsights, Deutsche Bank.

Geopolitics & Globalisation

Perhaps the biggest surprise was that people generally expect geopolitics to have a positive impact on their investments. There are some strong arguments for this, particularly as trade has proved resilient to the recent tariff wars. However, it seems that most people expect current geopolitical conflict to resolve in their country's favour.



Investors' anticipated impact of geopolitics & globalisation on their main investment portfolio over the next 3 years (0-10)



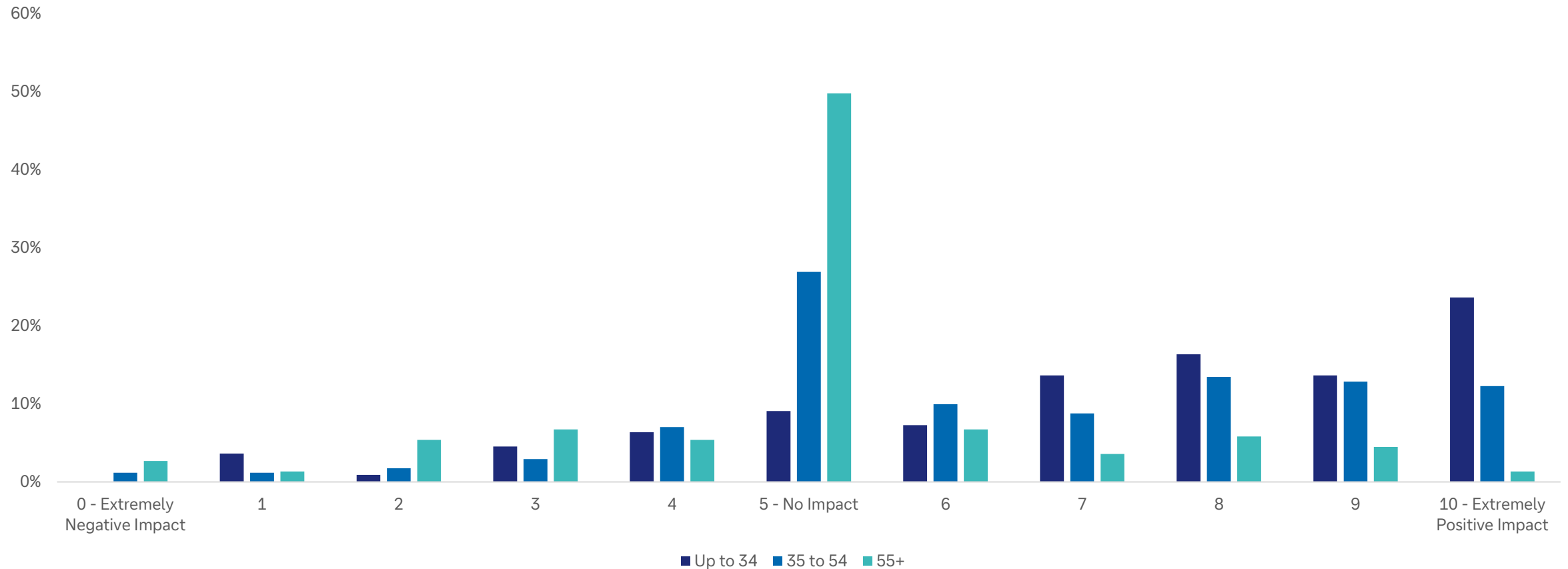
Source: dbDataInsights, Deutsche Bank.

Domestic politics & social unrest

Our model shows that domestic politics is heavily correlated with economic growth; however, investors do not rate it as a key investment factor.



Investors' anticipated impact of domestic politics & social unrest on their main investment portfolio over the next 3 years (0-10)



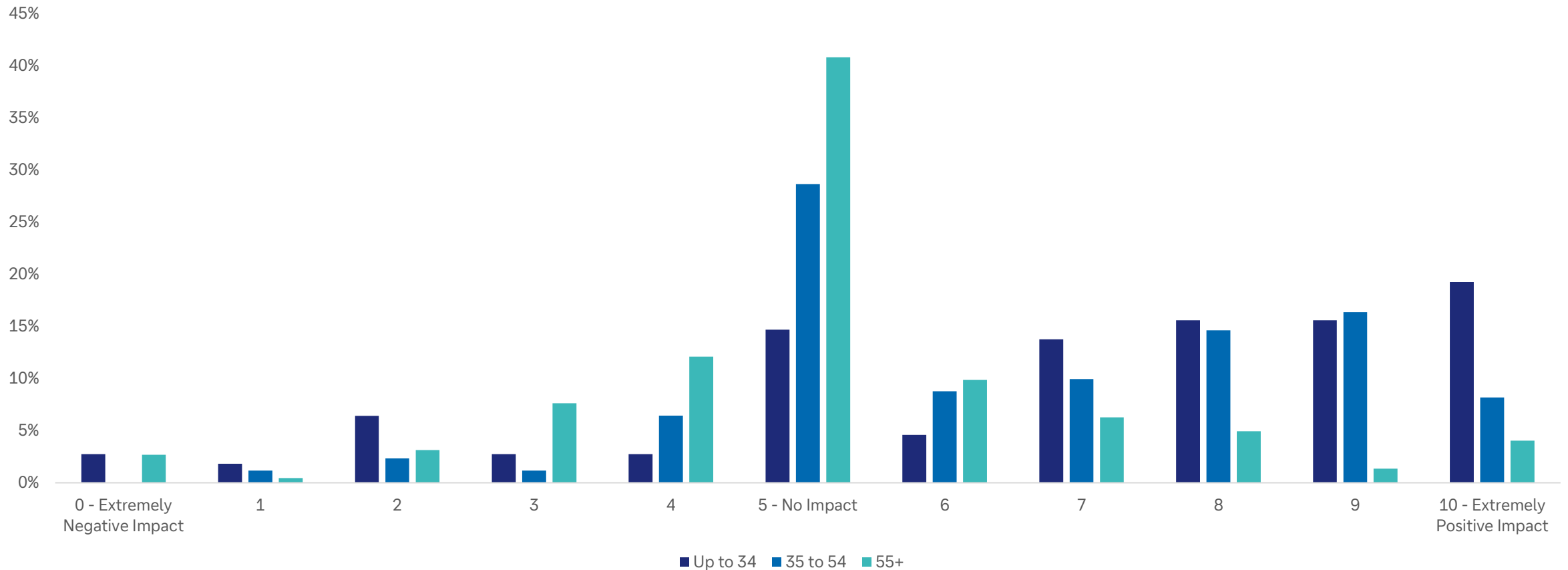
Source: dbDataInsights, Deutsche Bank.

Demography

Far more people are optimistic about demography. This also aligns with the positive tilt for sovereign deficits (which are deeply intertwined). To be fair, there are some underappreciated positive aspects to demography such as increasing healthspans and the consumer power of older people relative to those in the past.



Investors' anticipated impact of demography on their main investment portfolio over the next 3 years (0-10)



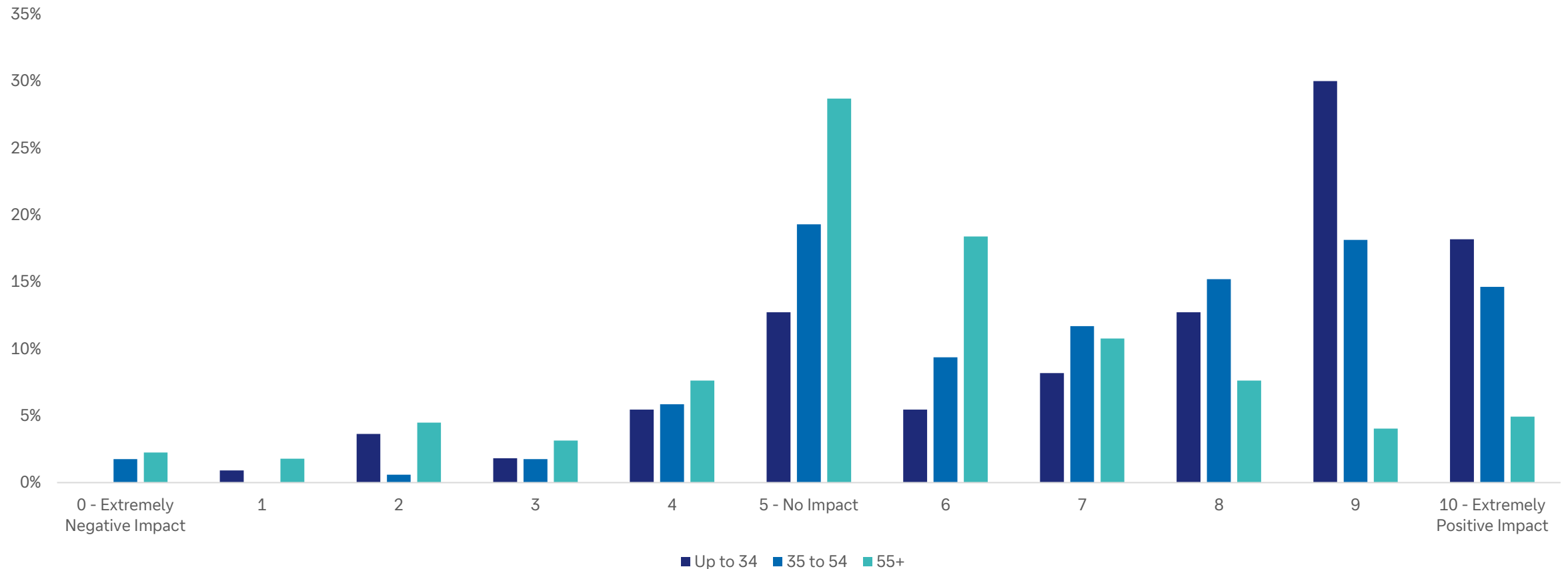
Source: dbDataInsights, Deutsche Bank.

Energy disruption & transition

There have been booms and busts in various alternative energy assets over the last decade. However, there is a strong realisation that energy (from all sources) will become increasingly important in an age of both AI and geopolitical weaponisation.



Investors' anticipated impact of the energy transition on their main investment portfolio over the next 3 years (0-10)



Source: dbDataInsights, Deutsche Bank.

Appendix 1



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Key takeaways from the Deutsche Bank Global Auto, Mobility & Robotics Conference

May 22, 2026

Key takeaways from the Deutsche Bank Global Auto, Mobility & Robotics Conference

Held at the Deutsche Bank Center in New York, the recent Global Autos, Mobility & Robotics Conference brought together industry leaders for two days of discussions on innovation, strategy, and the next generation of solutions and capabilities. The event provided a comprehensive view of how the sector is evolving – highlighting both near-term resilience and the structural shifts reshaping the industry. From changing demand dynamics and geopolitical pressures to the rise of new technologies and growth areas, the conversations underscore a sector in transition. These key conference themes capture the core insights shaping the outlook for autos, mobility, and robotics today.

Resilience for now, but risks building

Auto markets in Europe and the US remain relatively resilient, with demand holding up better than expected in the near term. However, visibility is limited and companies are increasingly cautious about the outlook. Geopolitical tensions – particularly in the Middle East – are a key risk, with the potential to push up costs and weigh on demand. While impacts so far have been contained, a more challenging environment is expected if pressures persist into 2027.

Costs rising, pricing power limited

Cost pressures are building across energy, raw materials, and logistics, and are starting to flow through supply chains. While companies have partly offset this through hedging and cost controls, pricing power remains limited – especially in the US and Europe. Rather than raising prices directly, firms are relying on product mix and model upgrades, creating ongoing tension between protecting margins and sustaining demand.

Geopolitics driving cost pressures

Direct exposure to the Middle East remains limited for most companies, but the conflict is beginning to impact costs – particularly in energy and logistics. While hedging has softened the near-term effect, pressures are expected to build into 2027 as protections roll off. Early signs of supply chain inflation are emerging, with higher costs gradually passed through the system, often with a lag, and companies stepping up efficiency measures to offset the impact on margins.

New growth areas emerging beyond autos

Suppliers are increasingly looking beyond automotive markets to drive future growth, targeting areas such as data centres, energy systems, and defence. While these segments remain small today, they are seen as important medium-term opportunities. Continued investment and partnerships are expected to gradually scale these businesses and diversify revenue streams over time.

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China reshaping global competition

Chinese automakers are rapidly expanding internationally, gaining share across Europe, Southeast Asia, and South America. This is reshaping competitive dynamics and forcing established players to adapt through localisation, pricing changes, and new partnerships. For suppliers, aligning with Chinese exporters presents growth opportunities, but also intensifies competition and pricing pressure in global markets.

EV momentum moderating

EV growth is continuing but at a slower pace, as consumer demand shifts toward hybrids and traditional vehicles. In response, companies are taking a more balanced approach to powertrain strategies and scaling back the pace of EV investment. This reflects a more pragmatic, demand-led approach, with hybrids benefiting from extended lifecycles and a broader mix of technologies gaining prominence.

Efficiency and restructuring a key focus

Companies are stepping up efforts to improve efficiency, including restructuring programmes, underutilized plant closures, and footprint reductions. These measures aim to protect margins in a more uncertain environment. At the same time, adoption of automation and AI is accelerating, as firms seek to streamline operations and build more flexible, cost-efficient business models.



Appendix 1

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ChatIPO: Quick take on OpenAI's record-breaking IPO plans

May 21, 2026

Putting a giant share sale in context

The news that OpenAI is preparing to file for an initial public offering very shortly suggests that the maker of ChatGPT aims to strike while the iron is hot.

The poster child for AI plans to make a confidential filing with regulators as soon as tomorrow and aims to go public as early as September, the Wall Street Journal [reported](#) late yesterday, citing people familiar with the matter.

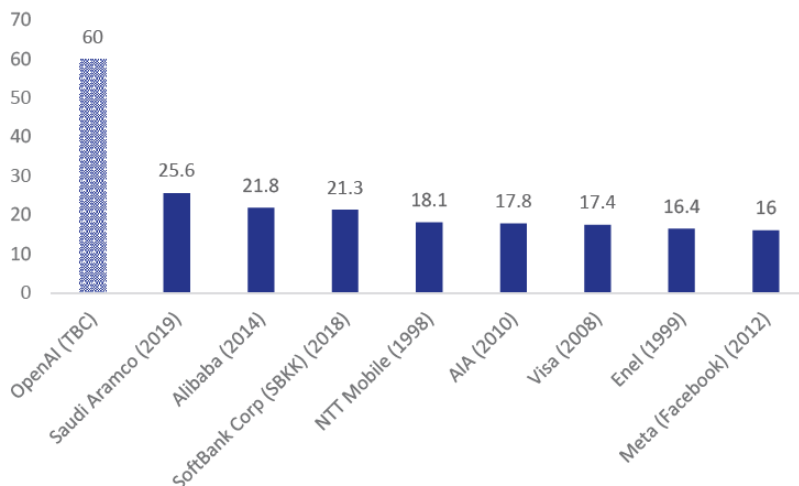
With an anticipated surge in IPOs this year, including of its fast-growing rival Anthropic, there will be a scramble to make the most of investor appetite for direct exposure to pure-play AI companies in the public markets.

Retail investors currently have to pick companies with AI offerings in other parts of the supply chain, either in infrastructure, such as semiconductor makers or cloud providers, or applications that sit on top of the foundation models. Shares in the Magnificent Seven tech companies have quadrupled since OpenAI launched ChatGPT on November 30, 2022.

OpenAI was valued at \$852bn after completing its latest funding round at the end of March. Reports have [suggested](#) it could be planning to raise \$60bn in an IPO that would value it at more than \$1trn. That would be the biggest IPO ever, twice as big as Saudi Aramco's IPO, which was valued at \$25.6bn in 2019, assuming no one else gets there first.

Even so, it has yet to be seen how public markets will value OpenAI and its peers once they open up their financial statements to scrutiny and explain the still little-understood economics of their business models.

Figure 1: A \$60bn IPO would be more than double Saudi Aramco's record (\$bn)



Source: Investopedia, media reports, Deutsche Bank Research

Authors

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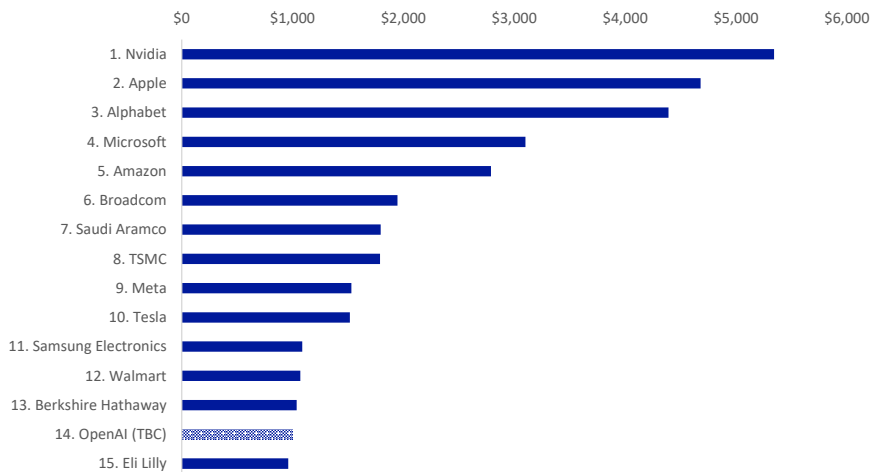


A trillion-dollar valuation would also make OpenAI, which has yet to make a profit and will [reportedly](#) make \$30bn in annualised revenue this month, the 14th biggest company in the world by market capitalisation.

That would be just behind Berkshire Hathaway, which made revenues of over \$370bn and net earnings attributable to shareholders of \$67bn last year, and just ahead of Eli Lilly, whose sales topped \$65bn and profit was \$21bn.

The most valuable company is Nvidia, perhaps the nearest thing investors currently have to a large pure-play AI investment, whose shares have surged more than 13 times since the launch of ChatGPT to give it a market capitalisation of \$5.4trn.

Figure 2: A \$1trn valuation would make OpenAI the world's 14th biggest company (\$bn)



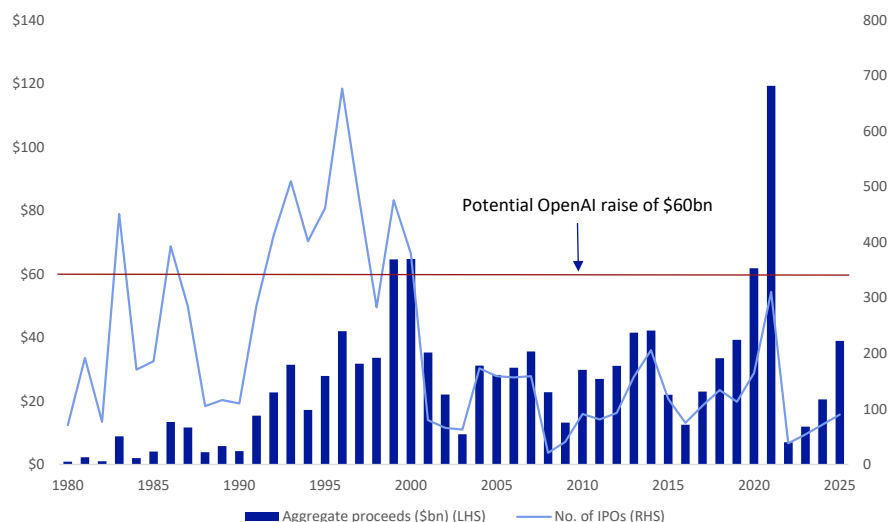
Source :Bloomberg Finance LP, Deutsche Bank Research

OpenAI is in a race with Anthropic, which overtook it in sales last month and is on track to generate \$40bn in annual recurring revenue this month, according to [data](#) from The Information. The maker of Claude may be looking to raise more than \$60bn in its own IPO this year, media reports [say](#), even while it is simultaneously in [talks](#) with investors about raising cash at a valuation of \$900bn – nudging it ahead of OpenAI.

A \$60bn IPO would also be larger on its own than the total amount raised in US IPOs in all but four years since 1980, according to data from the University of Florida. It would barely fall short of the \$65bn raised in both 1999 and 2000, and the \$62bn raised in 2020, and would be half of the record-breaking \$119bn raised in 2021.



Figure 3: A \$60bn IPO would be just short of three of the four biggest years for US IPOs since 1980



Source : Jay Ritter, University of Florida, Deutsche Bank Research

While there may be concerns about the capacity of the market to absorb a number of IPOs valued at several hundred billion dollars this year, they would slot into an US stock market worth about \$70trn overall. That is five times larger in nominal terms than it was even at the peak of the dot-com bubble in the late 1990s. At that time, there was an average of almost 500 IPOs a year, compared with about 120 this decade, and they are typically now coming to market at a more mature stage.

Last week, AI chipmaker Cerebras raised \$6.4bn, including the overallotment option, in the largest IPO this year. The shares jumped by two thirds on its first day of trading, giving it a market value of roughly \$67bn.

OpenAI has been preparing the groundwork for an IPO in recent months, streamlining its offering, focusing more on paying enterprise customers and clarifying its partnership with Microsoft. Earlier this week it won a long-running legal case brought by Elon Musk.

See the [Deutsche Bank Research Institute](#) for recent relevant research, including [Will Claude's AI finance agents change your life?](#) and [AI, Iran and the Gulf 101](#). The website is available to all so feel free to share.



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Megatrends: AI vs the decade's structural headwinds

May 20, 2026

Executive summary

We have never been more concerned about the six megatrends that impact economies and markets than we are today. Indeed, our new AI-powered megatrend model shows that the world faces severe headwinds from several overlapping megatrends that, post-WWII, has only been seen during the 1970s oil crises and the onset of the 2008 financial crisis.

In this piece, we explain our new megatrend model and how it uses AI analysis to quantify qualitative signals, filters our proprietary human-created data from dbDataInsights, and layers it all beside traditional data series. In all, we take almost 100 data points and track how each of the six global megatrends have waxed and waned quarterly over the last 70 years. In particular, we focus on the impact of these trends on GDP growth, equity markets and other key economic outcomes.

We then use our model to estimate how the most important megatrends will develop over the rest of the decade. We conclude that the defining feature of the next 5 years for developed economies will be the push-pull between technology and sovereign deficits. In other words, can the productivity benefits of AI technology outweigh the burden of growing sovereign deficits? All this will take place against the backdrop of several other megatrends which will both complement and work against us.

There is some good news. Our model shows that the current economic impact of technology is growing. And history shows that our technology indicator is correlated with growing productivity growth. Thus, how AI is adopted across corporates and economies will determine whether or not developed economies can overcome the drag of the megatrends working against us.

Finally, we look at the investment consequences of the current trajectory of the six megatrends. We outline a series of medium-term investment concepts that investors should consider, and then present a matrix that looks at several key asset classes. We also explain that traditional haven assets are not as effective as they were.

Megatrends are the biggest forces that shape markets and economies over time. Yet, they confound investors because they are rarely observable on any one day's trading. But what is clear from our model is that we are on the cusp of major historical change. We are certain that this transformation will demand more of us than has other episodes of profound shifts in the economic regime.

We thank the dbDataInsights team for their input into this publication.

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2. Interesting statistics on a page

- The other tech boom: In 2021, our technology indicator hit its highest point since the 1990s as lockdowns spurred massive tech investment.
- Through most of the 1960s, 70s, 80s, and 90s, technological improvements were associated with lower inflation. Although that relationship has reversed over the last decade, we put this down to non-tech factors (such as covid stimulus programmes). We expect greater tech improvement to lead to lower inflation in the medium term.
- Post-WWII, US real economic growth over any ten year period has averaged 3.8% pa.
- Since 1790, the US had 93 years where 10yr GDP growth was above 5% pa. In contrast, there were only 21 years where 10yr GDP growth was lower than 1.5% pa.
- 46% of Americans think today's investment environment is more certain than it was in the 2010s. Only 20% think it is more uncertain.
- 32% of Britons think today's investment environment is more certain than it was in the 1960s. Only 22% think it is more uncertain.
- 68% of the world's current GDP is generated by 12 countries in the G20 with declining working age populations.
- The role of domestic politics & social discontent may be underestimated by investors. Indeed, our megatrend indicator for this peaked in the 1990s - something that helped boost the tech boom. Today, it remains sharply negative.
- Support for populist left has surged over the last three years. In the 10 largest European countries, the populist left has similar support levels to centrist parties. The populist right enjoys the highest aggregate support.
- 67% of Americans believe the technology megatrend will help their investment portfolio over the next three years compared with 18% who think it will have a negative impact.
- Since 1990, the amount of GDP generated per kilogram of oil equivalent has quadrupled in \$PPP terms
- Americans are far more positive than are Britons about the impact of each of the six megatrends on their investment portfolio over the next three years.
- 45% of Americans think the trends in their domestic politics will have a positive impact on their investment portfolio over the next three years. In contrast, only 39% of Britons think the same.
- Indicators of domestic politics & social discontent tend to be well correlated with GDP growth.
- Over the last 20 years, M&A has become correlated with megatrends.
- Since 2020, the combination of six key haven assets have failed (positive correlation with equities) over more time periods than they have worked. In particular, havens failed to protect against risk downturns during four big shocks of the 2020s (covid, rate rises, tariffs, Iran).



3. A new model to better understand megatrends

The precursor to the internet, Arpanet, was created in the 1960s. But despite this system, and its successors, being well known for some time as academic and government tools, it took three decades before a sudden shift in the technology megatrend catalysed the commercialisation of the internet and caused the associated market boom in the 1990s.

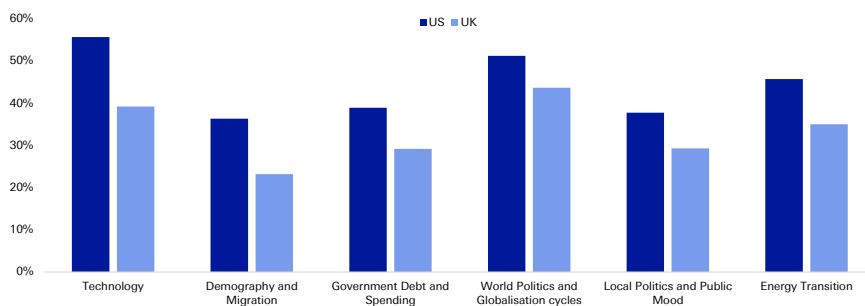
Megatrends are usually dinner party conversation because there is so much subjectivity and human bias involved. But now, AI can help us add the quantitative rigour needed to examine how megatrends affect markets and economies. Indeed, our megatrend model is only possible because AI allows us to quantify qualitative indicators, layer them alongside traditional data series and, most importantly, put an objective structure around the whole thing.

Megatrends have recently risen to prominence as our model shows that over the last 3 years (as interest rates have normalised) megatrends have become a key defining force in economies and markets. This has not been the case since before the 2008 financial crisis. As such, we expect them to define the rest of the decade and, as we will explain, we are deeply concerned at their current trajectory.

What investors think

Our recent dbdataInsights survey of investor attitudes towards megatrends shows that they are beginning to understand that megatrends have a growing influence on their portfolios. This one-off survey of US and UK investors complements our monthly surveys of a representative sample of the US and UK populations that feeds into our model. The following chart shows that investors in both the US and UK have made changes to their portfolios over the last three years as a result of all six megatrends.

Figure 1: Proportion of investors who have made a change to their investment portfolio in the last three years as a result of the following megatrends



Source : dbDataInsights, Deutsche Bank.

What exactly is a megatrend?

To us, a megatrend is a long-term, transformative force that shapes economies and markets. It can be seen in the data over multiple business cycles and has a proven ability to catalyse structural change. It alters how we live, work, interact, produce, and consume. It disrupts business models, create new industries, and reshapes



global power dynamics.

Unlike exogenous shocks, megatrends are predictable to a degree. While their exact trajectory and impact can be uncertain, the underlying forces driving megatrends are often identifiable and can be analysed.

What our megatrend model does differently

Unique human-created data is critical in AI analysis to avoid 'samey' output. Hence, we complement our qualitative signals with data from years' worth of our proprietary surveys. All this is layered alongside traditional datasets that track economies and markets. Taken together, we can then use AI to draw out themes by connecting datasets from different industries and macro groups.

The end result gives us a structured framework for analysing how the world's megatrends are changing and how they are impacting markets and economies – specifically: GDP growth, equity prices, inflation, and interest rates.

Megatrends impact society on a different timeline to their impact on companies, economies and markets

The time gap between when a megatrend is talked about and when it actually impacts markets and economies can be very wide. Electricity, railroads, the internet, and demography are all obvious examples. Investment decisions that are too early are wrong. Our model aims to fix that by addressing megatrends in terms of their observable impact on market and economies.

As such, different aspects of our model can be either: 1) leading indicators of economic and market impacts or, 2) corporate, market or economic indicators themselves.

"Megatrends are back" – after decades in the wilderness, they will likely be the driving factor to 2030 and beyond

In 2015, our 'geopolitics & globalisation' megatrend indicator dropped heavily. That came as the shortage index fell, trade policy uncertainty rose, and the Kilian index was low. Just 12 months later, the couplet of Brexit and Trump triggered the end of unfettered globalisation and upended the way economies and market see risk.

Over the last four decades, people have either ignored or oversimplified megatrends:

- 1990s: Investors assume the once-in-a-century tech boom was solely responsible for everything that happened (spoiler alert: it wasn't, as we discuss later).
- 2000s: people treat the worst financial crisis in 70 years as an event in-and-of-itself that caused widespread problems, rather than being caused/augmented/amplified by many other trends.
- 2010s: QE and low rates mean nothing else mattered.
- 2020s: The many shocks of Covid, Ukraine, tariffs, Iran etc. are seen as one-off exogenous events and not part of the normal megatrends shaping economies
- The point here is that investors have largely treated most of the last 40 years as either being driven by unique, unlucky events or by low interest rates. As a result, there is very little institutional memory about how megatrends



influence markets and economies in 'normal' times. Investors are simply not used to analysing and incorporating megatrends in their investment decisions.

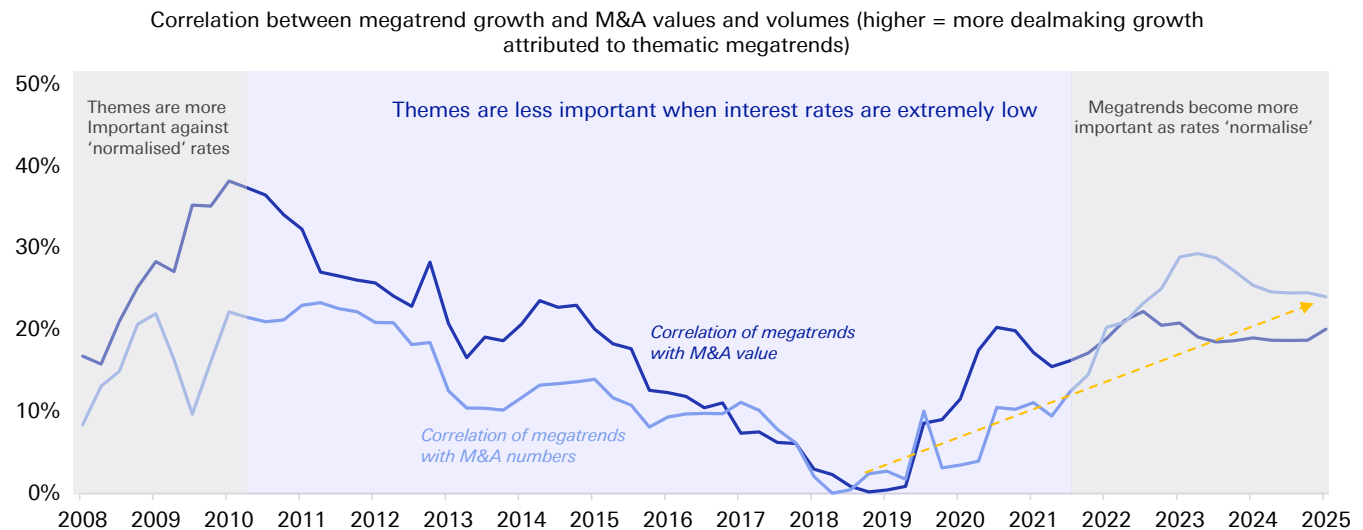
Now that we have moved out of the era of both very low interest rates and the covid aftermath, megatrends should have a more profound and obvious impact on markets and economies.

Greater volatility will amplify megatrends

Our megatrend indicators tell us to expect more volatility – in geopolitics, society, markets and, thus, corporates. Indeed, we have already seen how various sudden events this decade have led to supply shocks and these are increasingly driving an economic and market response – both negative and positive. On the negative side, they have pushed up inflation and constrained output. On the positive side, they have encouraged countries and corporates to pursue greater resilience.

This expectation of greater volatility comes right as the world settles into an era of more 'normal' interest rates. This is critical as low interest rates (and the aftermath of the aggressive rises in 2022) have been the defining feature of economies and markets for the last 15 years. Investors have become complacent as a result as the effects of low interest rates are sometimes very hard to see (our model shows it clearly). One of our favourite ways to see how interest rates have affected economies and markets is to compare megatrends to the thing that neatly intersects both these things – M&A.

Figure 2: M&A has become increasingly correlated with the growing impact of megatrends



Source : Dealogic, Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank. * Note: rolling 10yr R^2

Investors still underestimate how normalised interest rates are propelling megatrends to the forefront of economies and markets. Our analysis shows that everyone should pay more attention.



4. The megatrends and their impact on economies and markets

"There is no reason for any individual to have a computer in their home," – so goes the infamous quote supposedly uttered in 1977 by the founder of Digital Equipment Corporation. That same year, Apple unveiled the Apple II computer and the rest is history ...

Our megatrend model is based on the analysis of almost 100 data series (including AI quantification of qualitative indicators, and human-created survey data). These have helped us create a quarterly indicator for each of the six megatrends that shows how it positively or negatively impacts markets and economies over time.

Below are the six megatrends that our analysis indicates should impact markets and economies. We summarise our view on each individually in the appendix.

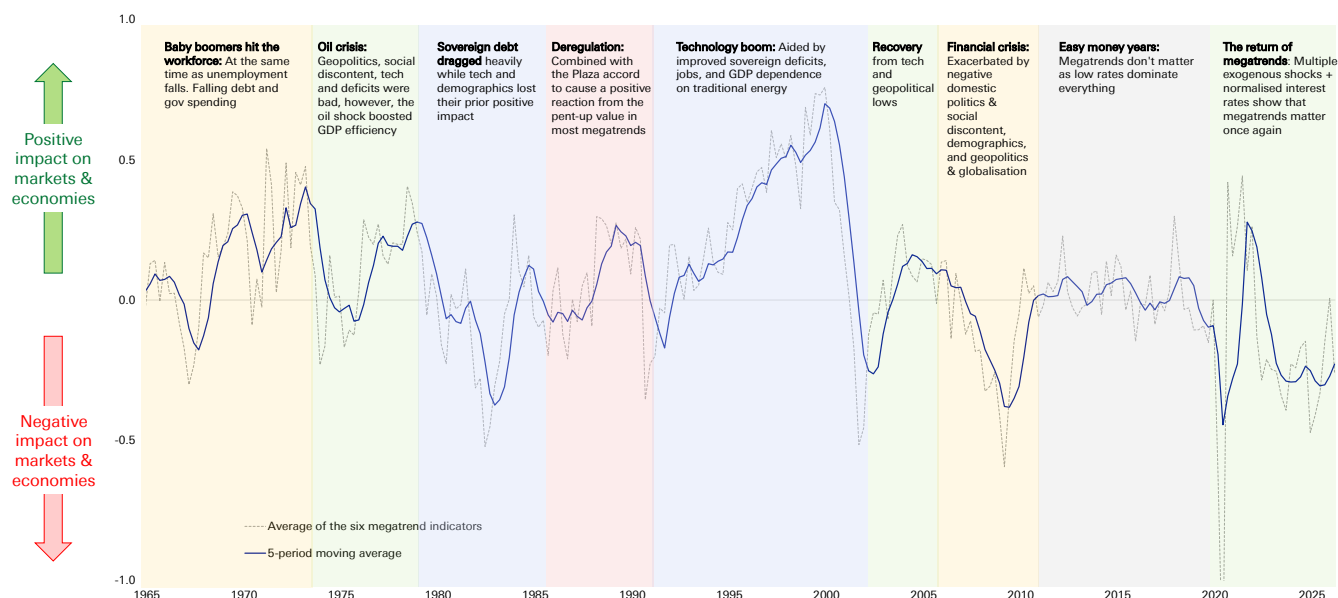
- **Technology** – The 'adoption gap' is crucial to our analysis. This is the time period between when a technology is invented and when it has an actual impact on productivity, GDP etc. Our indicators predict that AI will turbocharge productivity and push markets and economies to new highs.
- **Sovereign debt** – This is the most important negative megatrend. The worsening deficit situation is being exacerbated by demographic decline. Yet, we are cognisant that some periods of high debt in the past have been followed by periods of high growth when other megatrends align.
- **Geopolitics & globalisation** – There are many negative signals, however, deteriorating international relations does not necessarily mean a linear negative impact on economies. Trade in goods can be maintained via reorganised supply chains. Arguably more important is the trade in ideas across borders which is permanently accelerating due to ever greater and quicker connectivity.
- **Domestic politics & social discontent** – Rising discontent is causing voters to support more extreme politicians with policies that may address short-term issues at the expense of longer-term macro prosperity. This trend will continue to drag on economies and markets
- **Demography** – This goes beyond an ageing population and immigration. Our indicators show that other factors are very important including, the incentive for young people to form households, have children, and participate in the economy.
- **Energy disruption & transition** – Our most complex megatrend: Energy security and diversification are key. However, we also see that although energy disruption causes near-term pain, it can lead to long-term positive effects. From this point of view, rising prices can incentivise processes that are more productive per unit of energy.



The current impact of the six megatrends on markets and economies

Our series of almost 100 datapoints eventually aggregate into one measure for each megatrend (standardised via z-scores) and our model examines these quarterly for almost the last 70 years. We can then aggregate these six megatrends into one indicator to see how the collective impact of megatrends overall is impacting markets and economies. The following chart shows that when our indicator is positive it means that megatrends, in aggregate, are having a positive impact on economies and markets. When it is negative, the reverse is true.

Figure 3: The combined impact of megatrends on economies and markets



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

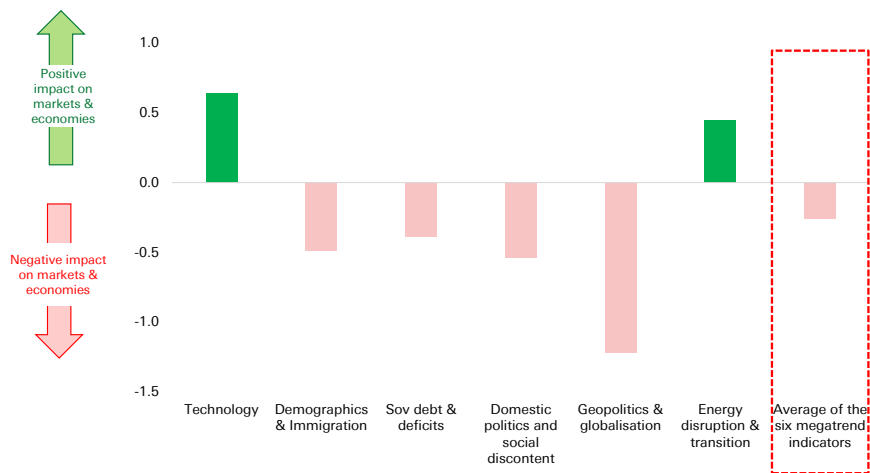
There are many important takeaways from the above chart. Just a few include: 1) Today, the world faces severe pressure from the negative effects of megatrends, 2) The impact of megatrends on markets and economies can turn quickly, even if the underlying trend moves slowly, 3) Note the period during the 2010s (the era of low interest rates) where megatrends didn't matter.

Which megatrends are having a positive or negative impact on the world?

The following chart shows the position of our current megatrend indicators as of the latest quarter. Of course, any given quarter can have its own volatility, which is why we use a rolling average in the above aggregated chart.



Figure 4: The impact of megatrends on economies and markets (at March 2026)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

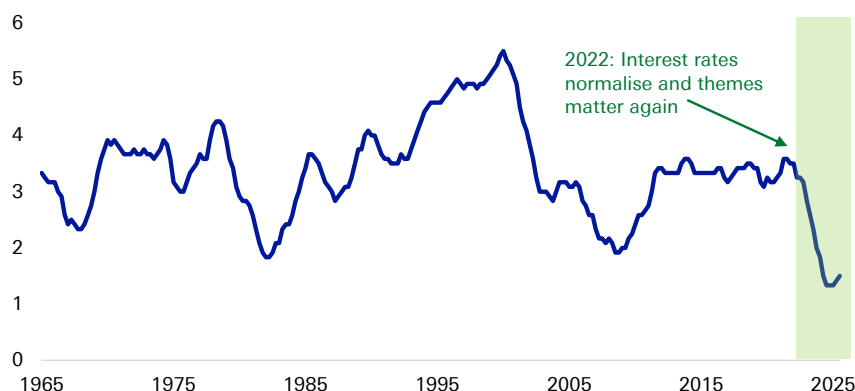


5. The big concern today

Our aggregate megatrend indicator ([Figure 3](#)) tends to be more positive than negative. Indeed, it has spent twice as much time in positive territory than negative territory.

Yet, today, the indicator is deeply negative. Adding to that concern is that the number of megatrends working positively is historically low – this can be seen in the following chart. The two similar occurrences of these lows were associated with severe downturns (note that these figures are based on a 12-quarter moving average in order to minimise the noise from quarterly events).

Figure 5: Number of megatrends (out of 6) that are having a positive influence on markets and economies (12qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

Megatrends are not independent from each other ...

A typical flaw in many megatrend analyses is that they address the various themes independently because the overlap between them is just 'too hard' to figure out. Our model adjusts for this by accepting there is overlap between some of the data series that impact each theme. In addition, we standardise all the results so that the six megatrends can be compared for their impact and then aggregated. As such we, as investors, can see how important any combination of themes is at any point in time.

... nor are they consistent

Megatrends do not have a constant impact on the world, and they do not grow at a steady rate. Rather, their impact on economies and markets waxes and wanes over time. The following example shows how.

An example: Why the 1990s were so good

[Figure 3](#) categorises each of the eras above by the 'typical' reason cited for them. However, this is far too simple. That is because different megatrends work in different phases – at times they complement each other, at other times they work against each other.

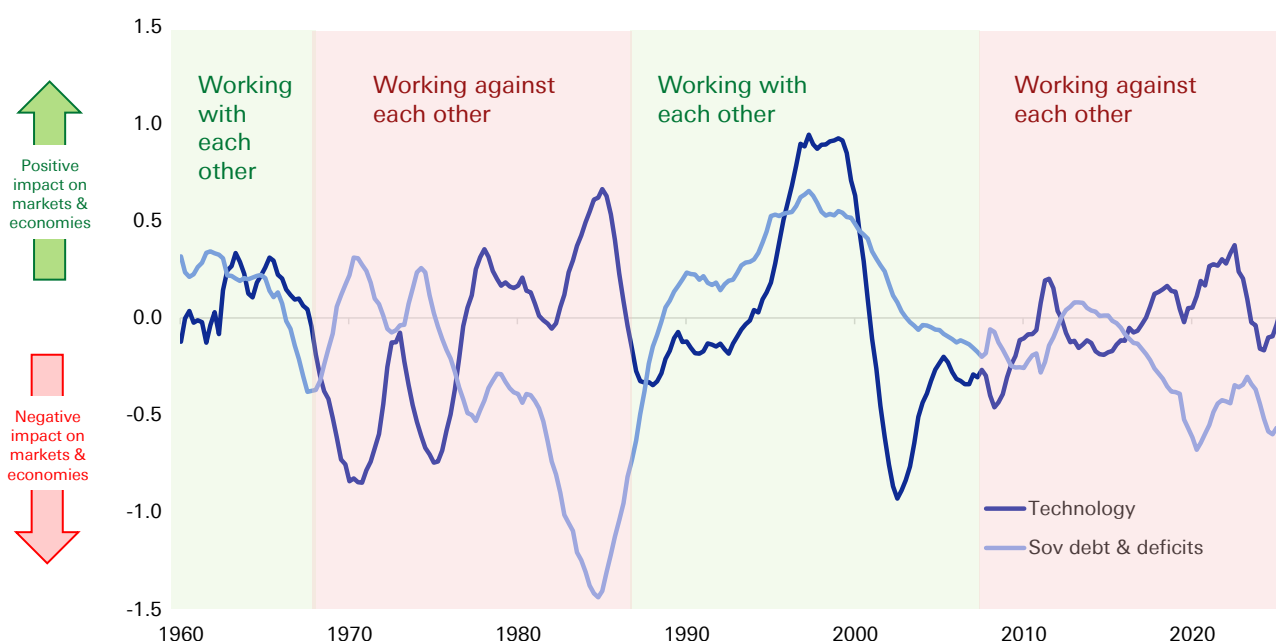
Take the 1990s. This decade was such a good period for markets and economies not just because of the tech boom. It was also a period where most of the megatrends



were positive most of the time. Thus, relatively stable politics aided policy, the workforce was generally strong, and the background of all this reduced concerns about sovereign debt. The combination of several strong megatrends provided the fuel for the technology boom.

To illustrate, let's show just two of the trends. The following chart shows the interaction between the technology megatrend and the sovereign deficits megatrend. The green shading shows the periods when the two have worked together and the red shading shows the periods where the two have worked against each other.

Figure 6: Megatrend indicators for 'technology' and 'sovereign deficits' (12q average)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

The implication for the 1990s should be clear. Technology was racing ahead with the proliferation of the internet at the same time that growth in US government spending was low or negative, debt was falling as a proportion of GDP, and the 10-year term premium was falling. The combined result turbocharged markets.

In other words, the positive impact of technology was amplified by the positive effect of the improving sovereign fiscal position. And that is before considering that three other megatrends were also making a positive contribution (geopolitics & globalisation, domestic politics & social discontent, and the energy transition). In fact, only the demographic megatrend worked against things. This was driven by rising house prices and falling growth in household numbers, which more than offset the positive economic impact of rising immigration.

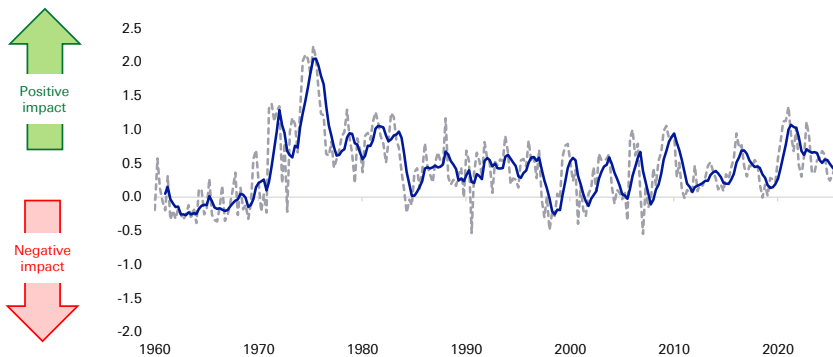
In an equal and opposite way, the tech bust in the early 2000s did not impact the world by itself. Indeed, the bursting of the tech bubble coincided with 9/11, war, and recession. It was a time when the housing prices accelerated, the job market was tough, real wages ran negative, and the labour participation rate fell.



Another example: Lessons from the 1970s oil crises

The oil crises were overwhelmingly bad for society and the economy. However, during the 1970s, our megatrend indicator showed a mixed response. That is because the negatives of geopolitics, social discontent, technology and sovereign debt were offset somewhat by a huge increase in the energy megatrend. As a result, the world started finding new ways to create energy away from the dependence on the Middle East. Since 1990, the amount of GDP generated by one kilogram of oil equivalent has quadrupled in \$PPP terms (World Bank). The chart below shows how the energy disruption megatrend has remained broadly positive over time.

Figure 7: The impact of energy disruption & transition on markets and economies (trendline is 5qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

Today's tech boom may be different from that in the 1990s

The background to today's technology boom is very different from that of the 1990s. Whereas the 1990s version was complemented by the positive effects of most of the other five megatrends, today's is not. Most megatrends are working against economies and markets. Therefore, if a technology boom is going to help bail out the malaise in the world economy it may have to work even harder to overpower the other megatrends.

So, is it possible for AI to be significantly more impactful than was the growth of the internet in the 1990s? Our analysis shows that not only is it possible but also it is the most likely outcome. This is extremely fortunate for the global economy as several other megatrends are working against the world and need to be overcome in order for us to avoid a slip into economic malaise.

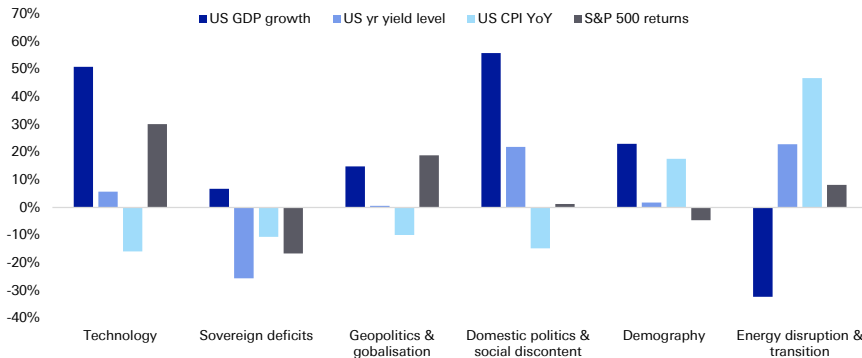
How important are our megatrend indicators for markets and economies?

The main point to tracking the six key megatrends is to assess their impact on key markets and economic variables. The following table shows the correlation of the megatrends and four key factors: US GDP growth, the S&P 500, 10yr treasury yields, and US interest rates. These calculations start, at the latest, from 1962 and in most cases earlier depending on data availability.

The following chart shows the 'all-time' (since 1960) correlation between our megatrend indicators and the four economic and market factors.



Figure 8: Full sample correlations between our megatrend indicators and the four economic and market factors (Q1 1960 to Q1 2026)



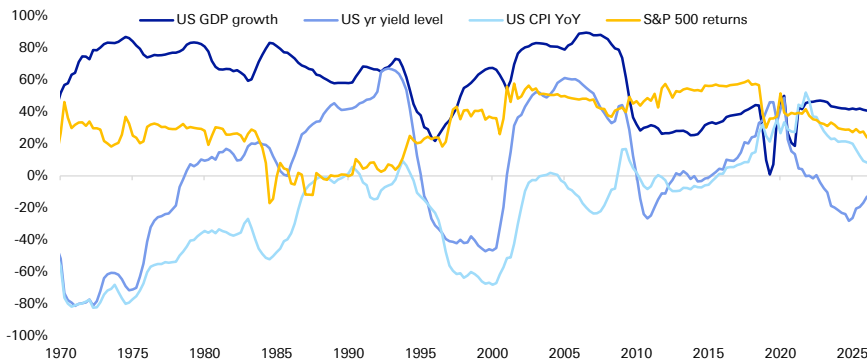
Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

There are two points to note

1. Big numbers – each of the four market and economic variables is materially correlated with several megatrends.
2. In any given period, a megatrend may be more or less correlated with the four market and economic variables than the above ‘all-time’ table indicates.

Take, for example, technology. This is generally very important for GDP throughout all periods. However, its effect on the other three variables has alternated between being very important and less important. The following chart shows how this has changed over time. In particular, the overall correlation with 10-year treasury yields is low, however, it is wrong to dismiss the relationship – as the following chart shows, technology and treasuries have had a very important relationship throughout most of the last 60 years.

Figure 9: Rolling 10yr correlations of our technology megatrend indicator with the big four indicators



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.



6. Where our model shows megatrends are taking the world

The status quo is almost certainly wrong

Many corporates and investors today say they are preparing for volatility but, actually, they assume the status quo. They assume average economic growth of 1-3%, ongoing stability in debt markets, inflation that is elevated but manageable, and average stock market returns.

This status quo is almost certainly wrong. Our megatrend model shows that, as we look to 2030 and beyond, one of two scenarios for the developed world is likely, and it all depends on technology, in particular AI. Indeed, the backdrop of deteriorating sovereign debt and the demographic overhang means that technology is the most important, most likely (and perhaps only) way that the world can fight and win against this and other negative megatrends.

We must be honest and say that AI can go one of two ways – either it will fulfil its promise (as we believe) and turbocharge productivity, or it will fizzle into an average tech with merely incremental impact (we exclude the tail risk that AI goes rogue and destroys the world). In other words, either sovereign debt and poor demographics will become an overwhelming burden, or, technology will overcome it and push economies and markets to new highs.

So, investors and corporates should prepare for either a productivity boom or a severe, prolonged downturn. And regardless of which scenario wins, because AI is now at the beginning of the J-curve of adoption, the coming few years will be punctuated by upheaval.

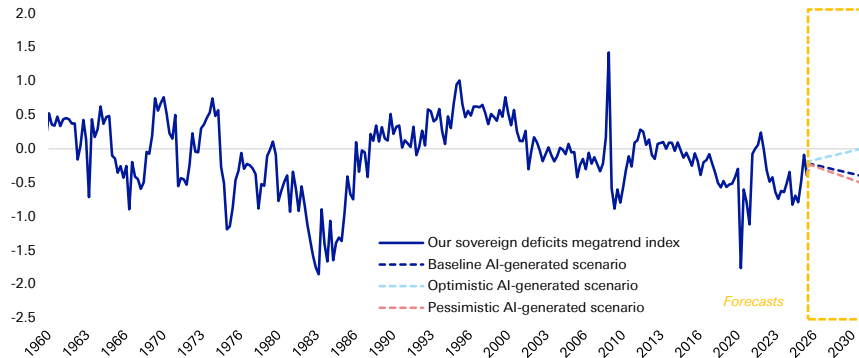
What tech is fighting against

Of all the six megatrends, the most worrying is sovereign deficits. In many developed countries, the pressure of already-large debt piles is likely to grow as the projection for budget deficits remains deeply concerning. Indeed, US debt is expected to rise from about 100% of GDP to 120% by 2036 (CBO).

The following chart shows how our model expects the sovereign debt megatrend to deteriorate over the rest of the decade.



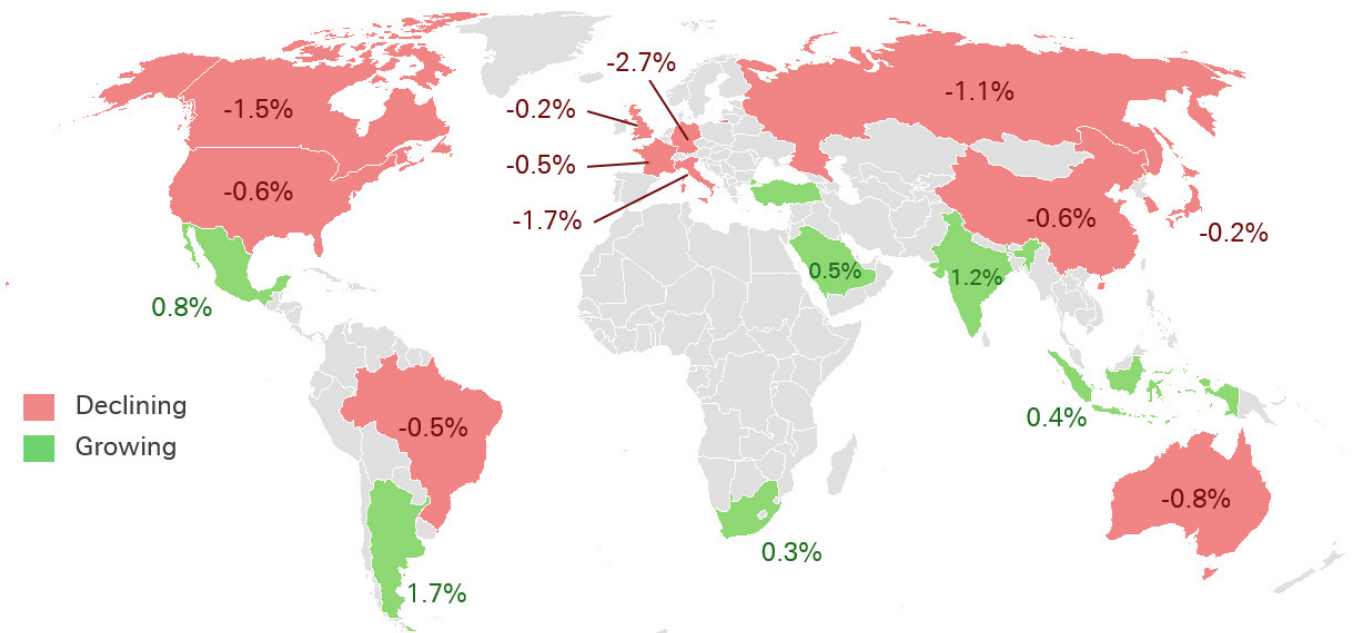
Figure 10: Our sovereign deficits megatrend index with AI-generated scenarios to 2030



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

A key driver behind the sovereign debt problem for most developed countries is poor demographics. The following chart shows the extent of the problem.

Figure 11: Growth in G20 countries' working age (20-64) population between 2025 and 2030 (as a % of the total population)

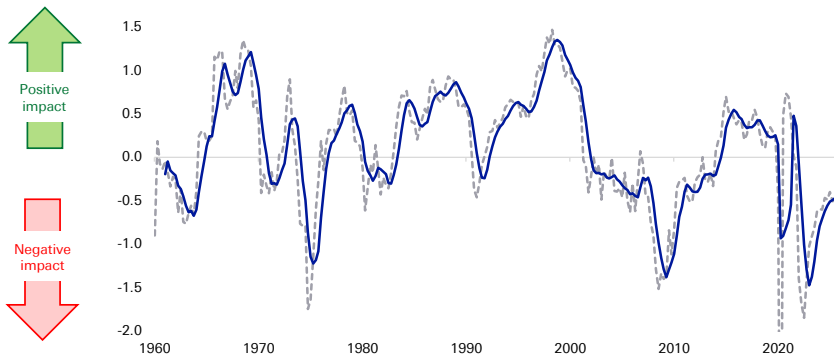


Source : UN, Deutsche Bank.

Demographic problems are well known. What is less well understood is how the megatrend of domestic politics and social discontent may also worsen the sovereign debt megatrend. Indeed, we measure social discontent based on many measures including, housing affordability, real wage growth, labour share of income and more.



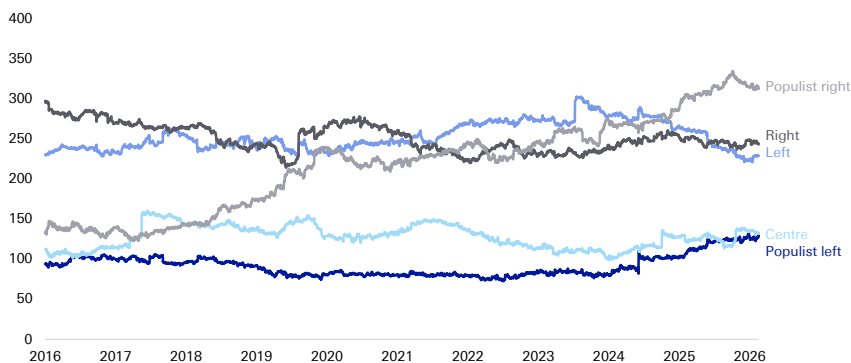
Figure 12: Our domestic politics & social discontent megatrend index



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Growing social discontent raises the risk that people will vote for populist politicians (from any side of politics) that may enact policies that add to debt today without regard for the long-term consequences. Of course, some politicians that are described as “populist” can exact positive, constructive policies, however, we fear that many typically do not. The following chart shows how populist support in Europe has increased over the last decade.

Figure 13: Sum of support for each category of current political parties across the 10 largest European countries



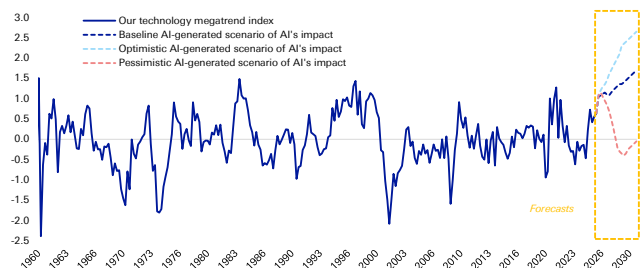
Source : Politico, dbDataInsights, Deutsche Bank.

Can tech outweigh the worsening trend of sovereign debt?

Yes, it can but the impact of AI on individuals, corporates, and economies has to accelerate from here. Our model shows this is the most likely outcome per the left-hand chart below.

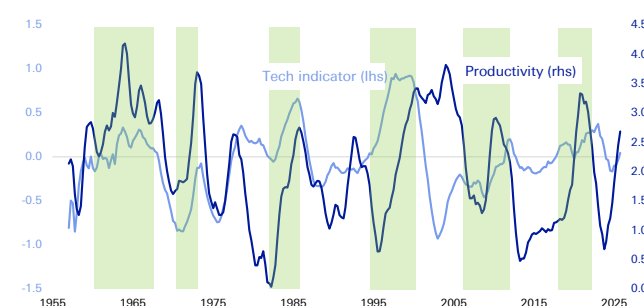


Figure 14: Our technology megatrend index with AI-generated scenarios to 2030



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Figure 15: Technology megatrend indicator v productivity growth (12qtr moving avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

The sharp turn up in our technology megatrend indicator is encouraging because this indicator, over time, has given us a reasonable idea of the direction of productivity growth – the key thing that will drive economies and markets forward. In general, whenever our technology indicator has accelerated to a peak, productivity growth has accelerated sharply. Notably, in the 1990s, it accelerated to 3-4% for consistent periods of time. There one big exception – in the late 1970s during the oil crisis and heady geopolitical tensions.

Our baseline expectation (per the above left chart) is that by 2030, our technology indicator will accelerate beyond levels seen during the 1990s technology boom. That implies we may see sustained productivity growth even greater than the 3-4% seen during prior peaks of our tech megatrend indicator.

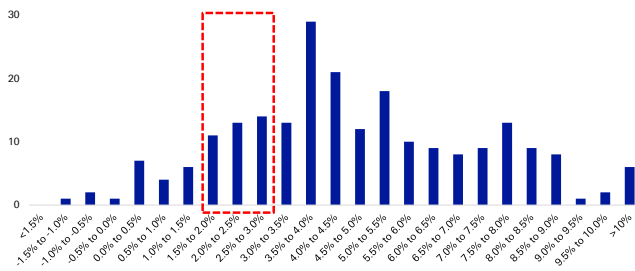
History tells us that a bad sovereign debt position can be overcome

The “Long Depression” in the US in the mid-1870s came just a decade after the Civil War and, more immediately, followed a financial crisis and investor fear at the overbuilding of railroads. A lack of government intervention exacerbated the economic malaise and unemployment likely topped double digits. The sense of despondency was palpable. However, towards the end of the decade, the benefits of the railway technology began to boost the economy. At the same time, several other trends in agriculture and sovereign debt moved in a positive direction and resulted in startling economic growth.

When we look at the history of the US economy, big jumps in growth are not unusual. In fact, they are quite prevalent. The following charts break down US economic growth into buckets. The red box surrounds the range that we usually expect GDP growth to sit in. But even these are infrequent compared with the jumps we have experienced.

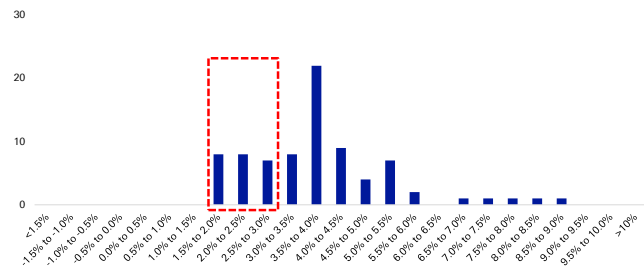


Figure 16: Frequency of US GDP growth (1790-present, rolling 10yr, annualised)



Source : Finaeon, Deutsche Bank.

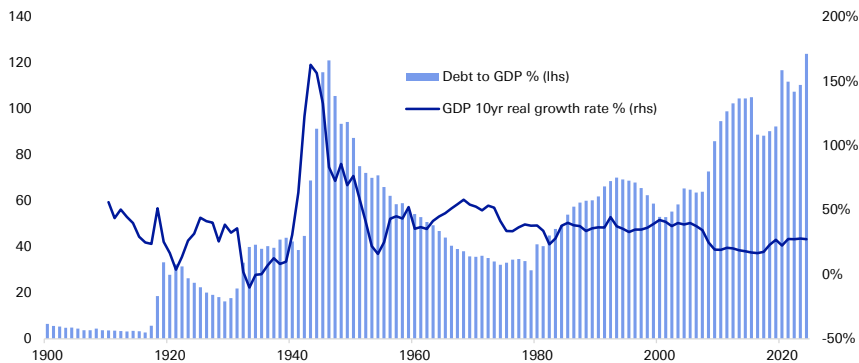
Figure 17: Frequency of US GDP growth (post-WWII, rolling 10yr, annualised)



Source : Finaeon, Deutsche Bank.

There are some critical instances through history where high sovereign debt levels were followed by periods of strong economic growth. One of the most notable is in the US after the second world war. At the war's conclusion, the US embarked on a 35-year period of deleveraging. And yet, this was a period of relatively strong economic growth – as the chart below shows.

Figure 18: US debt-to-GDP and GDP growth rate



Source : Finaeon, Deutsche Bank. (Note: Current figures differ from the CBO figures, however, this longer-term data series is used here to show the trend)

A key reason why economic growth was strong in the decades after the Second World War, despite high debt levels, was that other megatrends were working in a positive direction. Notably, globalisation grew quickly, domestic politics were reasonable, social discontent was low, and there was a positive impact from energy technologies.

Today, most megatrends show worrying signs which is why we are almost entirely dependent on AI to bail out the world economy. So where will the US, and other developed economies, end up over the coming ten years? Given the wide variety of outcomes from AI, it is unlikely that we will see the 'normal' growth rate of 2-3% pa. Our model suggests we are more likely to see higher growth with a lesser probability of low, stagnant growth. It all depends on the megatrends and particularly technology.

Some cause to be optimistic in the near term

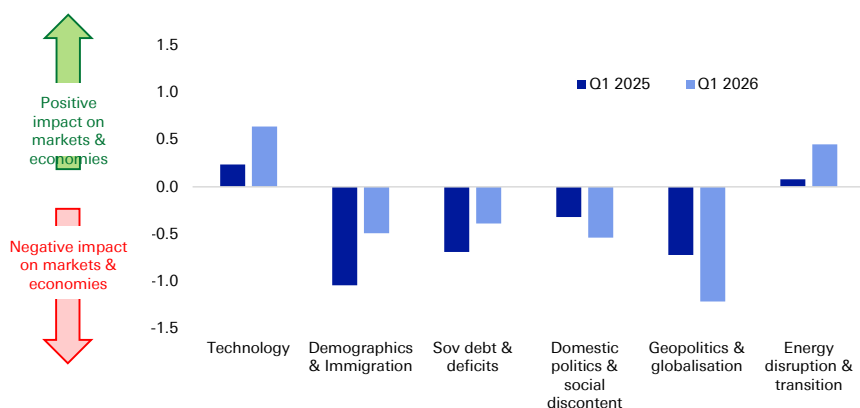
We do not put too much weight on the quarter-to-quarter movements in our megatrend indicators – there is frequently a lot of noise. However, we are conscious that the overall indicator has recently stabilised after its heavy drops over the last few years. This has been helped by a surge in the indicators for technology (driven



by AI applications and data centre investment) and the energy transition (driven by a surge in renewables and commodity prices that have incentivised a lower dependence on energy for GDP growth).

In addition, the following chart shows that four of the six megatrend indicators have improved over the last 12 months.

Figure 19: Change in megatrend indicators over the last 12 months



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

When we take a step back, our megatrend indicator is positive in 60% of the quarters since 1956 (the beginning of our data). And when it is positive it tends to have a strong effect. In fact, the total positive impact of megatrends is double the negative impact.

What should corporates do?

Despite the volatility ahead, corporates must not let themselves be dissuaded from investing heavily in AI and other platforms for their business (whether physical or intangible). That is because as the volatility settles over the next few years, the AI-powered productivity boom may very well send economies to new highs. To take advantage, corporates must buy or build the right tech, physical assets, and organisational structure.

There has never been a better time for corporates to invest in AI and related infrastructure. Corporates in the S&P 500 alone are sitting on over \$1.2tn of cash on their balance sheets while European companies in the Stoxx 600 sit on \$1tn. Meanwhile, private capital has over \$2tn of dry powder, so there is almost \$5tn sitting in institutional money market funds. And that is before considering that debt financing options remain cheap and easily accessible.



7. Investment consequences

As our megatrend framework helps us look into the medium term, we can use it to identify some specific trends that we believe will help frame investment decisions today:

- The volatility caused by exogenous shocks in the future will be greater than the volatility from these shocks in the past.
- Stockpickers should benefit as stock prices move more in line with company fundamentals.
- Continued decoupling of the US and China despite sporadic pauses for 'strategic stability'.
- Unpredictable sovereign debt sell-offs – particularly triggered by exogenous shocks and investor emotion.
- Haven assets could become risk assets – treasuries and the dollar, in particular, will become more volatile if the sovereign deficits megatrend deteriorates further.
- Private markets may become more attractive as investors look to avoid volatility.
- Uncorrelated assets should likely become increasingly popular (and expensive) as volatility and risk hedges (such as music rights, sports assets etc).
- Hedging will be difficult if haven assets lose their reliability.
- US over RoW based on:
 - Strong customer culture
 - Tech and capital markets leadership, and the institutional depth to support both
 - Demography is stronger than other DMs + China
 - Other countries now have to allocate resources to develop things that were formerly underwritten by the US, such as defence.
- Advantaged companies typically have:
 - A defendable moat rather than efficient supply chains
 - Natural geopolitical and currency hedges
 - Prioritise asset efficiency over size
- Capex-heavy business models could become just as attractive as asset-lite businesses
- A releveraging cycle should benefit companies
- Companies will tend to pursue transformational M&A rather than the 'hopeful' M&A they implemented during the era of low interest rates
- Return on equity will likely be more important than profit growth
- Companies may boost returns by spinning-off underperforming divisions
- We are cautious on markets where state-run enterprises play an important role



- Stocks with higher credit quality to outperform whilst the refinancing needs of highly-levered stocks should be closely watched

The following table sets out more specific views based on the current direction of megatrends:

Figure 20: Our medium-term thematic views by asset class

Asset class / Horizon	3+ years
Equities	Long US stocks in (1) knowledge-intensive industries like TMT and healthcare and (2) cyclical consumer industries; EU countries with stronger growth and less concern about sovereign debt to outperform; European firms with strong know-how to relatively outperform; US and US-aligned stock markets to outperform non-aligned markets; negative on oil and gas stocks; defence stocks to outperform; equities performance across countries and cyclicals v defensives to align along demographic dynamics; healthcare to benefit most clearly
Credit	Trade-exposed materials and industrial companies to face margin pressures from tariffs and trade realignment; this could widen HY vs IG gap; Spreads to normalise further in both the US and Europe but relatively more so in Europe; Spread widening will be back-stopped by excess liquidity
Yields	US yields will remain among least negatively affected major economies when it comes to inflation and debt concerns; Sovereign yields of countries with ageing population and no credible fiscal plan for greater dependency ratios to come under volatility; otherwise lower yields in countries where ageing is more "natural" (EM)
Commodities	Polarisation of commodities / trade along axes of influence will benefit producers in choke-point countries where diversification is impractical; economic policy, demographics and tech to be a net negative for oil prices and energy stocks; renewable energy will grow but associated stocks may struggle. A wave of public-to-private deals is possible
Alternatives	Private capital (PC) as a major beneficiary of greater concerns about aging population and pension shortfalls; PC will be key in financing strategically important industries given likely lack of red tape cutting as well as budgetary constraints in many major countries. This will be especially true in Europe (energy, infrastructure); A cycle turn for private credit will be coming; but private capital's momentum is unlikely to stop; it will become ubiquitous in institutional portfolios, with a rise in allocations likely

Source : Deutsche Bank.

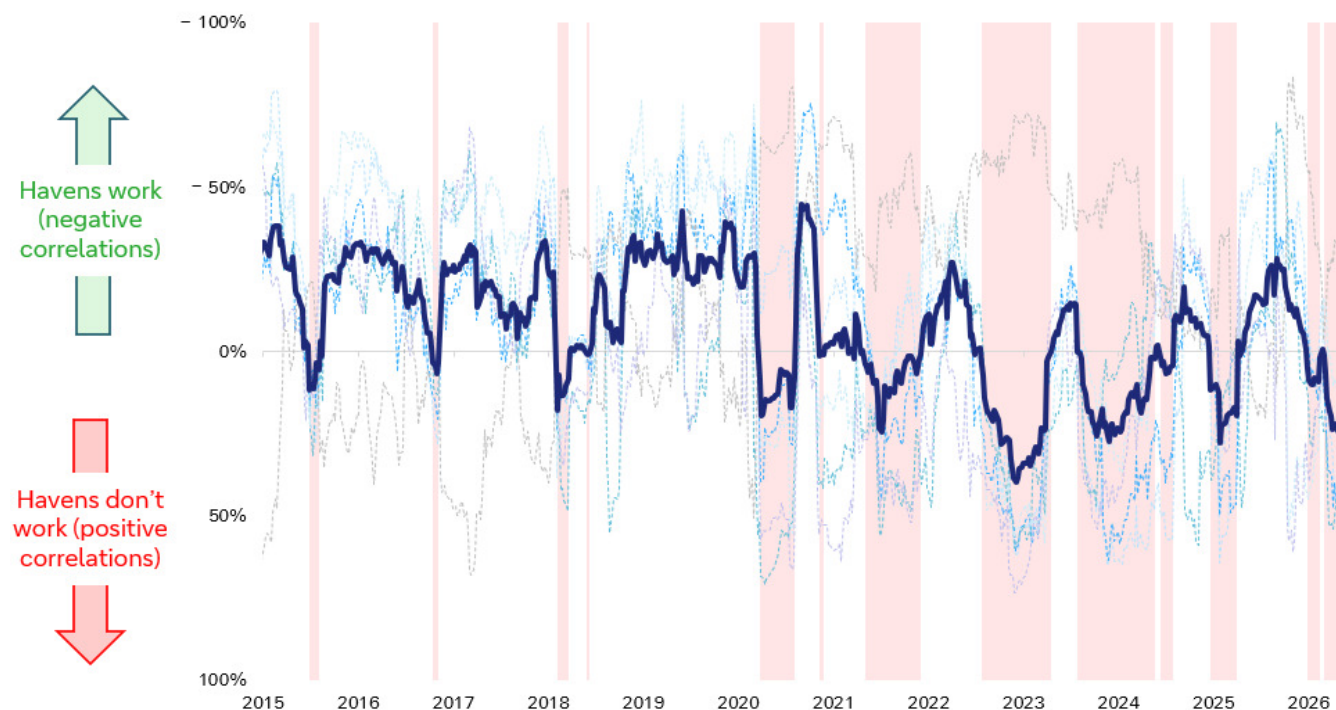
What about the haven assets?

Understanding megatrends is important because normal hedges may not work anymore.

As we think about the potential market downsides of megatrends the obvious question arises: "What haven assets should I buy?". The worrying thing is that, in recent times, the answer is increasingly difficult to ascertain. The reason is that many of the traditional haven assets simply do not work well anymore. To examine this, we looked at six haven assets: Gold, the US dollar, the Swiss Franc, the Japanese yen, 10-year treasuries and 10-year bunds. We then aggregated their performance when is shown on the following chart:



Figure 21: Our haven indicator (20-wk rolling, avg across pairwise correlations with S&P 500)



Source : Bloomberg Finance LP, Deutsche Bank.

The most important takeaway from this is that haven assets have not worked during the most important times, ie. Major risk-off moments, including Covid in 2020, the interest rate hikes of 2022, the US tariff announcements in 2025, and the Iran war in 2026.

This breakdown in havens was highlighted by the IMF earlier this year. It warned that this effect may “amplify systemic vulnerabilities” due to forced deleveraging. That is a particular concern as central banks are reducing their bond holdings and the greater supply is being absorbed by price-sensitive private investors.



8. Appendix: Each megatrend in brief

Technology

Overall view

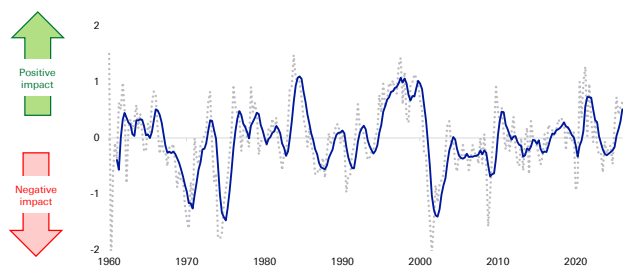
AI has been distributed and adopted far more rapidly than any other technology in human history. It has very low barriers to adoption and its applications will directly drive productivity for corporates. In the next phase of AI, institutions like firms, schools, health care, public administration, are adapting their organisation and processes to better accommodate AI. In some ways, this is analogous to the building of better roads to accommodate more cars in the early part of the 20th century.

The third phase will likely start in the early 2030s. Here, genAI systems will likely interconnect, share knowledge, self-improve exponentially, work across multiple domains. They will likely become partially self-aware and autonomous.

Investor perceptions

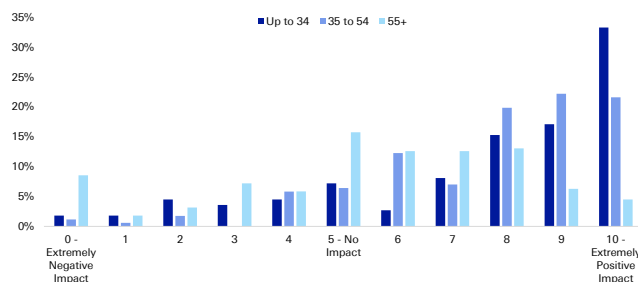
The technology megatrend is strongly correlated with GDP and equity prices. It also tends to push down inflation a little. The impact of technology on markets and economies has recently been driven by the huge investment in information processing equipment, along with its contribution to GDP. This is coupled with the continued investment in software (in its broadest definition).

Figure 22: The impact of the technology megatrend on economies and markets (trendline is 5qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Figure 23: Investors' anticipated impact of technology on their main investment portfolio over the next 3 years (0-10) by age group



Source : dbDataInsights, Deutsche Bank.

The above right chart shows investors are very excited about the potential for technology to positively impact their investments.

Three key technology variables to watch

1. Investment in information processing equipment: Growth has been extremely strong over the last few years as the AI boom has catalysed capex spending. A plateau (or drop) in spending would signal some near-term cooling.
2. Capacity utilisation for Hi-tech industries: This is the greatest level of output a plant can maintain within the framework of a realistic work



schedule. This has fallen from 88% just before the release of ChatGPT to 71% now. We look for increased capacity to signal greater AI adoption.

3. Industrial production of computers, comms, & semis: Growth here was poor several years ago but over the last 18 months has been trending up. Sustained growth will indicate greater domestic diffusion of technology capabilities.



Sovereign deficits

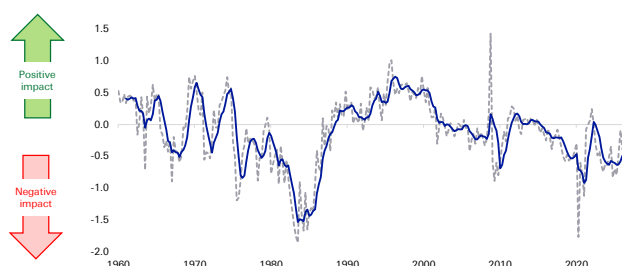
Overall view

In the era of normalised interest rates, we expect fiscal deficits, debt, and fiscal austerity to become contentious political topics. Should deficits continue on their current trajectory, we expect sustained levels of inflation above target and frequent, sovereign debt sell-offs. Associated fear would ripple through equity and credit markets. We do not expect a major sovereign default by 2030.

Investor perceptions

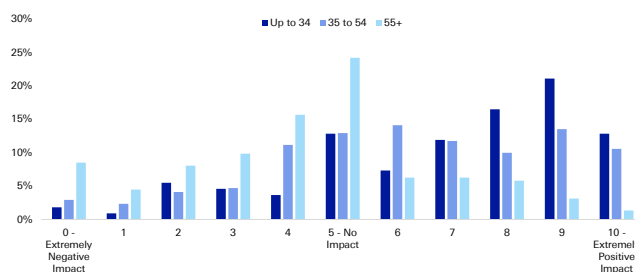
The impact of sovereign deficits on markets and economies has varied significantly over time. The below-right chart shows that investors are tending somewhat positive about the potential for sovereign deficits to impact their investments over the next three years. This may be because expansionary fiscal policy can boost markets over that timeframe.

Figure 24: The impact of the sovereign deficits megatrend on economies and markets (trendline is 5qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Figure 25: Investors' anticipated impact of sovereign deficits on their main investment portfolio over the next 3 years (0-10) by age group



Source : dbDataInsights, Deutsche Bank.

Looking out further, it becomes more serious. Indeed, the megatrend indicator (above left chart) has been trending down for the last three decades and the ageing population will apply further negative pressure. Indeed, US debt is expected to rise from about 100% of GDP to 120% by 2036 (CBO).

Three key sovereign deficit variables to watch

1. The contribution of government spending on GDP: This is a potential bright spot as this indicator has recently shot up. Although it is early days, if the continued trend here combines with greater AI adoption, there is potential for sovereign deficits to become less of a drag on markets and economies.
2. US 10yr yield vs. the G7 average: Treasury yields have been higher than the G7 average for most of this century. However, recently that gap has begun to shrink. Should this trend continue, it will signal that global demand for treasuries remains strong which, in turn, will boost the resilience of the US' twin deficits.
3. Fiscal policy uncertainty index: This turned deeply negative in 2024 and 2025, pushing below the Covid lows. While this is a somewhat subjective index, it is an important trend indicator because it can impact the short-term market view, particularly during an exogenous shock which can lead to fear-based sell-offs.



Geopolitics & globalisation

Overall view

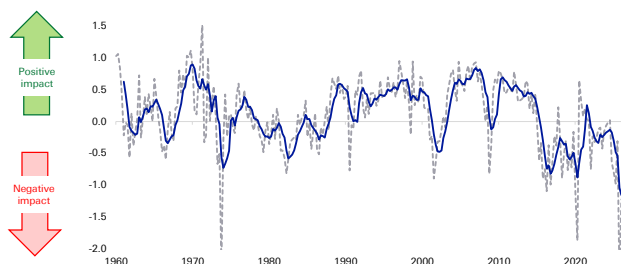
There are many myths associated with this megatrend. We believe globalisation and international trade will, in aggregate, continue to be strong, but it will reorganise itself around the US and Chinese spheres of influence. The US and China will likely continue to decouple their dependence on each other but the shocks of the 2020s show that corporates are very good at reorganising supply chains.

Many commentators have discussed how geopolitical shocks tend to only have a short-term impact on markets. Our megatrend indicator shows this is not true if you consider the broader picture and wider economy. That is because G&G impacts many things that are two, three, or more steps removed from the initial impact – and these may be incorporated into economies and markets slowly and over time.

Investor perceptions

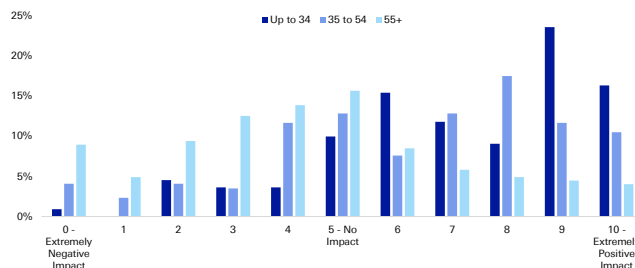
Investors are relatively nonplussed with G&G in terms of the potential impact on their investment portfolio. The below right chart shows that most regard it as neutral, although there is a leaning, particularly among young people, to be positive about the potential benefits of G&G.

Figure 26: The impact of geopolitics and globalisation megatrend on economies and markets (trendline is 5qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

Figure 27: Investors' anticipated impact of geopolitics & globalisation on their main investment portfolio over the next 3 years (0-10) by age group



Source : dbDataInsights, Deutsche Bank.

Our G&G indicator is somewhat positively correlated with the S&P 500 and GDP growth, and there is a small negative correlation with inflation.

Three key geopolitics & globalisation variables to watch

1. Shortage index: This covers US shortages of industrial goods, energy, food and labour. The combined shocks of this decade have caused the index to average its highest level since the oil shocks of the 1970s and then, before that, WWII. To the extent this remains high, it will indicate a worsening G&G impact
2. Trade openness: This is essentially exports + imports as a proportion of GDP. This has not changed much over the last decade despite much geopolitical upheaval. Thus, in itself, we see this as a positive indicator.
3. Kilian index of global activity: This is derived from a panel of dollar-denominated global bulk dry cargo shipping rates and may be viewed as a proxy for the volume of shipping in global industrial commodity markets



Domestic politics & social discontent

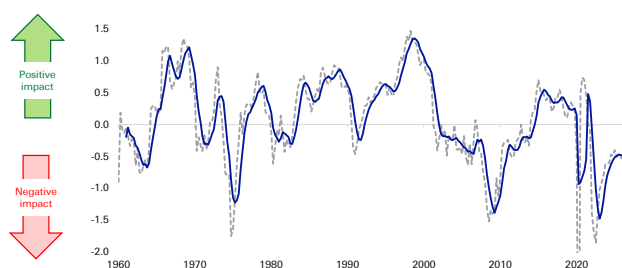
Overall view

We are recently past the 'tipping point' for populism (of all political colours). As such, it will likely become more widespread as the populist left grows to compete with the already-strong populist right. These politicians tend to encourage and amplify anger in voters. This would increasingly impact markets and economies as some countries potentially enact policies with negative long-term implications.

Investor perceptions

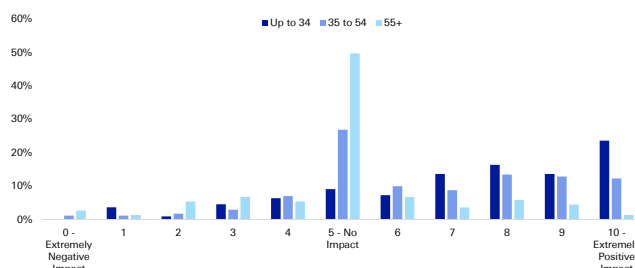
The way investors see domestic politics is similar to the way they view its sister trend, geopolitics & globalisation. Most investors tend to be neutral on the prospects for it to influence their portfolio, with a small proportion leaning positive.

Figure 28: The impact of domestic politics & social discontent megatrend on economies and markets (trendline is 5qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Figure 29: Investors' anticipated impact of domestic politics & social discontent on their main investment portfolio over the next 3 years (0-10) by age group



Source : dbDataInsights, Deutsche Bank.

Our megatrend indicator has a very strong correlation with GDP growth. Positive politics and lower social discontent also coincide with lower inflation and bond yields. Yet, we find that many people ignore the impact of domestic politics on markets as it is less impactful on equity markets.

Three key domestic politics variables to watch

1. Consumer sentiment: This gauge has done poorly at predicting what people usually assume – retail sales. But it has been good at estimating the level of social discontent. That directly translates into support for populist parties (on both the left and right) and can increase the likelihood that governments will embark on fiscal and institutional policies that have negative medium-term effects
2. Housing affordability ratio: For almost this entire decade, this indicator has been negative, indicating that affordability is growing worse. Most developed countries are stuck with the problem of incentivising more house building at the same time that regulation and local objections hold up construction.
3. Labour share of income: While growing real wages is positive, people care about their relative position in society. That is why the labour share is so critical. Labour share has been steadily trending down for decades and AI risks worsening this. This is one of those issues which typically lies dormant until there is an exogenous catalyst. When the frustration surfaces, various populist politicians may channel that anger in an unproductive direction.



Demography

Overall view

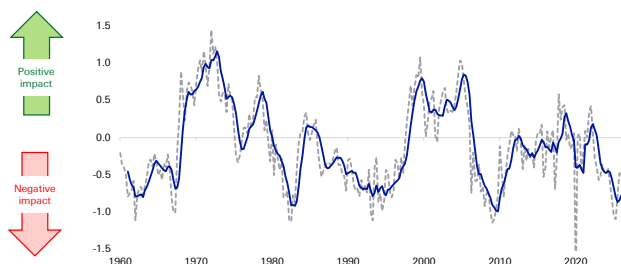
Demography is about far more than just the ageing population. People's 'health span' is rising, they are working longer and spending more. Still, pension savings are generally poor. That means people will take greater investment risk for longer, a positive for equities. Corporates will be forced to use technology to adapt to a tighter workforce.

Investor perception

The below left chart shows that the demography megatrend has deteriorated markedly since Covid. This is not only due to the declining working population but also falling growth in household numbers, lower net migration, and rising objection to migration.

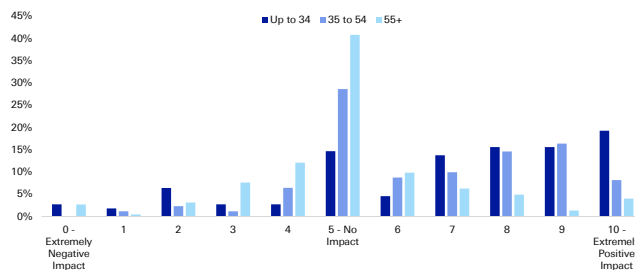
Younger investors err on the positive. They see demography as a positive for their investments over the coming three years. Partly, that could reflect optimism at how AI transformation is being accelerated by the need to address the declining workforce.

Figure 30: The impact of demography & immigration megatrend on economies and markets (trendline is 5qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Figure 31: Investors' anticipated impact of demography on their main investment portfolio over the next 3 years (0-10) by age group



Source : dbDataInsights, Deutsche Bank.

Three key demography variables to watch

1. **Net migration rate:** A big one for almost all developed countries where birth rates are below replacement levels. Politics will be a key driver here. That will be made even more pertinent if populist parties gain more traction in Europe. Some believe in closing borders, others believe in far more open borders.
2. **One-family house sales:** These are key as people are likely to have more children if they can buy a family home. The good news is that growth here has trended up over the last two years after a strong downswing in 2022 and 2023
3. **Number of households:** Household formation is generally a critical precursor to building a family (regardless of family home sales). The decade-long upswing in household growth abruptly ended with Covid. Since then, growth has been trending down. The overlapping social dynamics that caused this shift are complicated and poorly understood but this remains a key indicator for demography moving forward.



Energy disruption & transition

Overall view

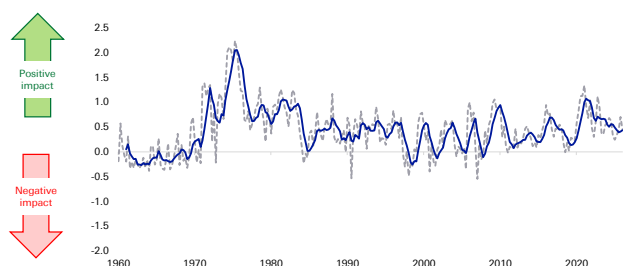
This megatrend is nuanced. Because we assess the medium term, we regard it as a positive when events happen that encourage greater energy security, diversification of supply, and adaptation that reduces the energy intensity of GDP.

The current energy crisis, therefore, can be a positive for economies and markets in the medium term as it is encouraging all the above points – even though the short-term effects are causing pain for consumers. In addition, the ongoing geopolitical tensions between the US and China is leading to a period of ‘strategic stability’ where countries are working to become more self-sufficient (or ‘friend-sufficient’) with critical minerals, commodities and other ‘chokepoint’ inputs.

Investor perception

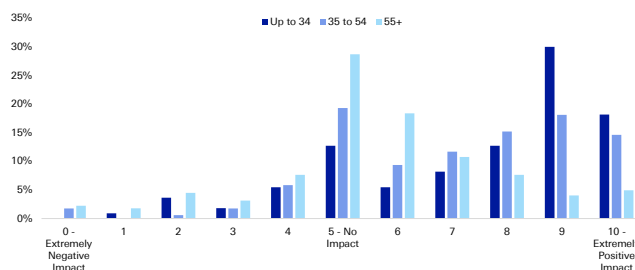
The good news is this megatrend has been positive for markets and economies for most of the last 50 years. That is because the world is gradually decreasing its dependence on traditional ‘chokepoint’ energy sources. Yet, there is a long way to go and the below right-hand chart shows that investors are generally positive about the prospect.

Figure 32: The impact of energy disruption & transition megatrend on economies and markets (trendline is 5qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

Figure 33: Investors' anticipated impact of the energy transition on their main investment portfolio over the next 3 years (0-10) by age group



Source : dbDataInsights, Deutsche Bank.

Three key energy disruption & transition variables to watch

1. **Metals price volatility:** Big spikes in various metal prices tend to cause panic in industry but, on the whole, they tend to revert to mean over time as either alternative supplies are sources, or alternative production methods are adopted. We expect commodities will be used as geopolitical weapons with increased frequency in the coming years, and we expect markets to respond in turn.
2. **Rig count:** There are several reasons why these may rise or fall over time. However, a falling rig count over the medium term is one indicator of the level of reliance on fossil fuels.
3. **Energy inflation:** In any one quarter or year, a rise or fall here can be noise, however, when energy inflation is falling over the medium term, it can indicate that supplies are becoming more diversified, stable and reliable.



Appendix 1

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How defence spending will shrink global imbalances and weaken US

May 12, 2026

In this note we argue that US withdrawal from security alliances established after the Second World War will result in a fall in global savings. This is as allies seek greater strategic autonomy in three key areas: defence, energy and technology. These areas are highly investment intensive and history suggests that when spending on these sectors rise, private consumption and investment do not fall sufficiently to offset the effect on savings. The result should be lower global savings and higher interest rates.

Because the regions most affected by a changing US relationship are also the world's largest net savers (in other words, they run the lion's share of global current account surpluses), the result should be a fall in global imbalances. There will be less recycling of savings out of Europe, Asia and the Middle East into the US and this will act as a material headwind for the dollar. For the sake of coherence, we focus in this note specifically on the defence angle but aim to build on this framework in the future when it comes to both energy infrastructure investment and technology.

The rapid global rise in defence expenditure

The recent conflict in Iran will accelerate an existing global trend: rising expenditure on defence. According to data from the SIPRI database, 2024 saw the largest rise in defence spending in real terms since data is available (the late 1980s - figure 1). Defence spending is set to rise over coming years, both in real terms and as a percentage of GDP. At the 2025 NATO summit in The Hague, member states committed to a 5% GDP spending target on defence and security-related spending by 2035.¹ Outside of NATO, Japan has committed to doubling its defence spending to 2% GDP by 2027, two years ahead of its original schedule. The Korean government has pledged to increase defence spending by 8.2% this year.² China has also been significantly increasing defence spending, by 7% in 2026 and Russian defence expenditure has grown dramatically since the start of its invasion of Ukraine.³

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¹ [NATO](#), April 2026

² [Yonhap news agency](#), October 2025

³ [Reuters](#) March 5th 2026



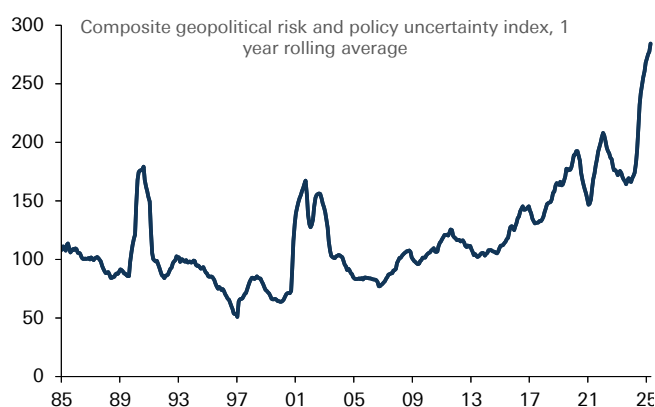
While the drivers of increasing defence spending are various (including rising nationalism, increased competition for raw materials), recent shifts in US policy have undoubtedly played an important role. The security umbrella established after the Second World War has come under strain from recent US policy decisions, including the withdrawal of military and financial support for Ukraine, tensions with EU allies over Greenland and most recently criticism of NATO allies over their decision not to back the recent US war against Iran. We have written in detail about increasing US reluctance to continue to provide 'global public goods' such as upholding an open international economic system and underpin western security, with the US pursuing a more self-interested policy agenda and exercising hard power and its economic networks more forcefully.⁴ This in turn is an important motivating factor for US allies to pursue strategic autonomy in defence, as well as other areas such as energy and technology.

Figure 1: Fastest increase in global defence spending in decades in 2024



Source : Deutsche Bank, SIPRI Defence Database

Figure 2: Reflecting extremely elevated levels of geopolitical risk and policy uncertainty



Source : Deutsche Bank, Caldara, Dario and Matteo Iacoviello (2022), "Measuring Geopolitical Risk," *American Economic Review*, April, 112(4), pp.1194-1225. Baker, Bloom and Davis.

Rising defence expenditure: falling global savings

To understand better what impact rising defence expenditure could have on the global economy, we turn to the historical record. We make three important observations.

First, never underestimate how quickly defence expenditure can increase, especially if threat perceptions change. Back in September last year, we [highlighted](#) the period of rearmament during the 1930s as a useful historical parallel. Defence expenditure rose dramatically across future belligerents in only a few years. For example, defence spending as a share of GDP between 1935 and 1938 rose 2.5% for Italy, 3.7% for the UK, 8% for Germany and 17% for Japan (figure 3).

Second, fiscal constraints can be overstated. Although Britain's debt to GDP stood at well above 160% in the mid 1930s, the country was still able to significantly ramp up defence expenditure. Arguably, the most basic function of a state is to protect its citizens and guard its sovereignty. As such, if threat perceptions increase, defence spending tends to trump other spending priorities.

⁴ See, [DB Special Report](#), "The Unruly hegemon; what a decline in US leadership means for markets," March 2025, [EM Blog](#): Venezuela, hard power, geoeconomics and hemispheric dominance, January 2026 and "[Geoeconomics 101](#)," January 2026

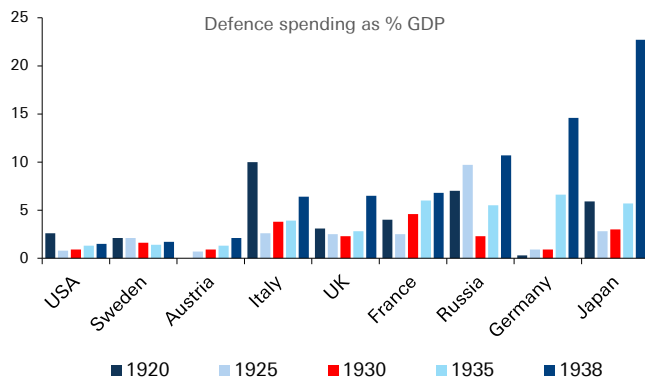


Third, rising defence expenditure tends to be associated with falling savings. From a national income perspective, defence expenditure can largely be accounted for through increased investment and government consumption. An important if not crucial question is whether this comes at the expense of private consumption (leaving the savings rate unchanged) or whether consumption does not fully compensate (reducing overall saving).

There is evidence to suggest that household consumption does fall during periods of re-armament. For example, in 1930s Nazi Germany, [economic historians](#) have described forms of moral suasion such as national savings days to increase available funding for the military, before the 1936 Four Year Plan saw rationing of raw materials and consumption goods which limited household spending opportunities. It is also possible that the uncertainty caused by a more volatile world prompts households to save for precautionary purposes.

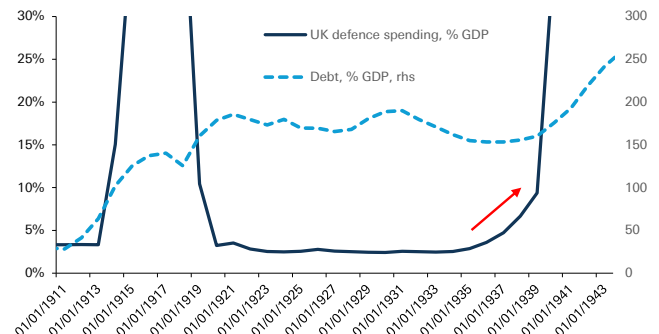
Nevertheless, efforts to increase national saving were never powerful enough to offset the demands of military investment at an aggregate level, reflected Germany's chronic current account deficit. And at the less extreme end of the spectrum, evidence also suggests that savings rates do fall. This was true in the UK during the late 1930s and the US following the start of the Vietnam War (figures 5 & 6).⁵ There is also strong theoretical evidence to suggest that defence spending is not subject to the same Ricardian constraints as other government spending. Government spending on defence does not crowd out other forms of spending because without it, the government is no longer able to protect the ability of households to save for the future or pass on bequests to future generations.⁶

Figure 3: Defence expenditure rose dramatically among belligerents in the run up to WW2



Source : Deutsche Bank, Eloranta, J. "Pro Bono Publico? Demand for Military Spending Between the World Wars," Economic and Business History Society, 2017

Figure 4: British defence spending rose in the 1930s in spite of very high debt levels

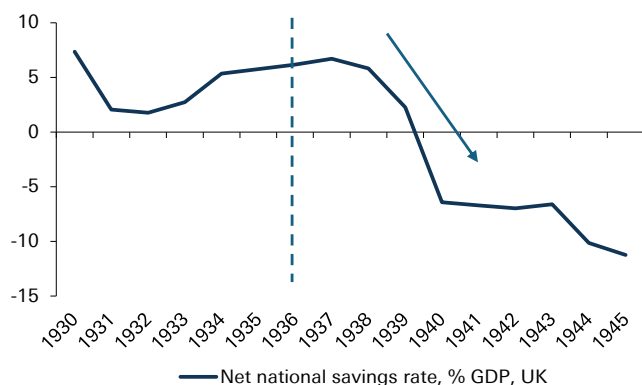


Source : Deutsche Bank, Bank of England Millennium of Macroeconomic Data

5 The UK savings rate does initially rise in the immediate aftermath of rearmament. However, this continued a trend established since the 1932 recession and rearmament did not truly start until 1937 after which national savings fell
6 See Seiglie, "Defence Spending in a Neo-Ricardian World." 1998



Figure 5: Savings rates initially rose then fell rapidly following re-armament in the 1930s



Source : Deutsche Bank, Bank of England Millennium of Macroeconomic Data, * dotted lines represent start of rearmament

Figure 6: They fell almost immediately in the Vietnam War



Source : Deutsche Bank, St Louis Fed, dotted lines represent start of rearmament

As we shall see, these observations have potentially highly important implications for global asset prices. We would note that while this report focuses specifically on defence, there are other sectors in which investment is likely to increase due to changing global political economy, especially energy infrastructure and technology. This should act to magnify the effect on global savings, even if the theoretical effect is clearest in the defence sector.

Falling global savings: shrinking global imbalances

Should increased defence expenditure globally act to increase investment without a corresponding fall in consumption, as we expect, the effect should be to raise interest rates, all being equal. In the long run, economic theory posits that the neutral rate of interest should be determined by the supply of and demand for savings.⁷

In the long run is an important caveat. In the short term, fiscal and monetary policy can also act to dampen the transmission channel. For example, during the Second World War, when British savings rates fell to very low levels, market interest rates remained extremely low. In part this was due to widespread financial repression policies, including capital controls, mandatory holdings of government debt and borrowing and consumer restrictions.⁸

We should be careful in taking the 1930s comparison too far when it comes to the current geopolitical environment. But governments will be confronted with hard choices when it comes to financing additional defence expenditure, especially where public debt to GDP is already at high levels, and given that welfare spending is considerably higher as a share of GDP than 90 years ago. The interplay between defence expenditure, interest rates and government policy is especially relevant when it comes to international capital flows, which we focus on next.

This is because in an era of capital mobility, another important consequence of rising defence expenditure will be its effect on global imbalances. The world's largest net savers (those running the largest current account surpluses) are

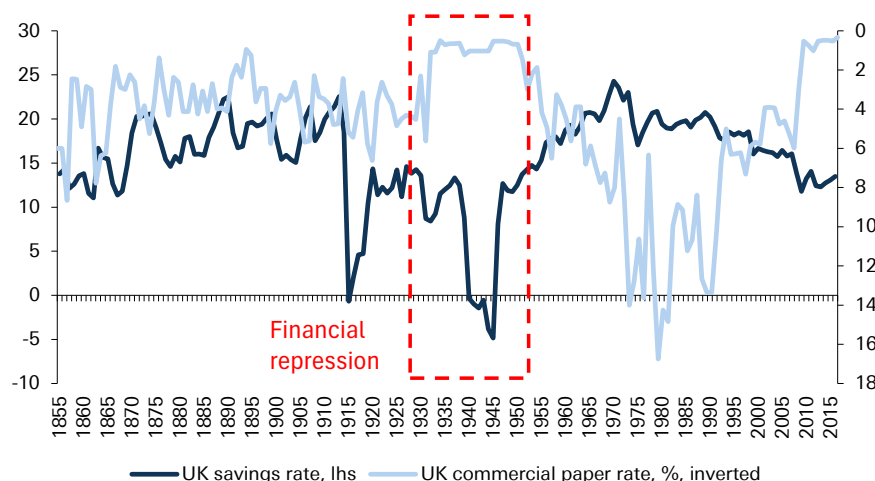
⁷ For example, Federal Reserve Bank of San Francisco [Economic Letter](#), April 2025

⁸ Mark Harrison, "[Paying the price for war](#)," University of Warwick



concentrated in three regions: Europe, East Asia and the Middle East (figure 9). These regions are also those seeing the largest increases in defence expenditure (figure 8), and where defence expenditure needs are likely to be the most pronounced over coming years.

Figure 7: In the Second World War, the government used financial repression to keep rates low despite collapsing savings



Source : Deutsche Bank, Bank of England Millennium of Macroeconomic Data

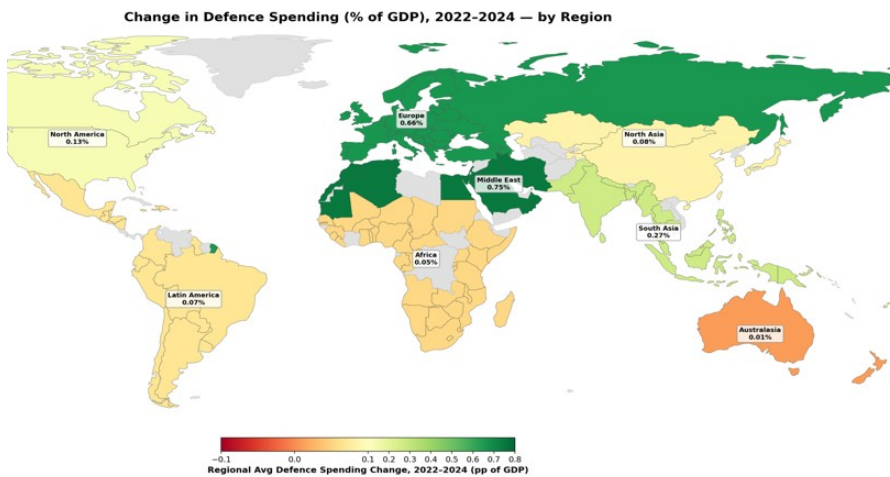
Increased defence expenditure will act to reduce global imbalances by reducing net savings in these three regions, in other words falling current account surpluses. But how will defence expenditure be financed? The quid pro quo of smaller current account surpluses should be reduced recycling of savings abroad via the financial account. From a political perspective, there are strong incentives to meet domestic financing needs from private savings rather than increased taxation or central bank financing. For one, it would be less politically unpopular than raising tax or inflation. For another, it reduces economic interdependencies which can make sovereign states vulnerable.

Indeed, as we noted in a [recent note](#), tapping private savings was a key channel of financing defence expenditure in World Wars 1 & 2. An even more extreme version of this could be outright repatriation of foreign assets to fund defence spending. Leading into the First World War, the economic historian Niall Ferguson has highlighted how international bond markets increasingly tilted towards 'home bias,' and during World War 2 Britain coerced private savers to swap dollar holdings in order to boost the Bank of England's foreign exchange reserves.

On the deficit side of global imbalances, as figure 9 shows, one country contributes more than any other combined, namely the United States. And it is also the case, when it comes to stocks, that the largest holders of US assets (when excluding the Caribbean holdings which are largely reflective of tax benefits rather than ultimate beneficial ownership) are once again: Europe, East Asia and the Middle East. In other words, if global imbalances fall due to increased defence spending, it is US deficits that may end up being the most vulnerable.

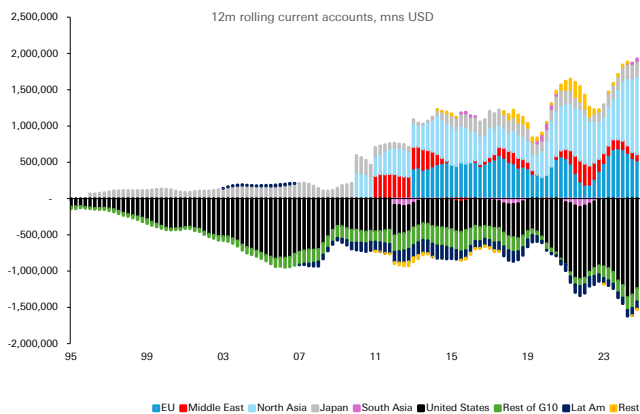


Figure 8: Europe, North Asia and the Middle East are seeing the biggest increases in defence expenditure



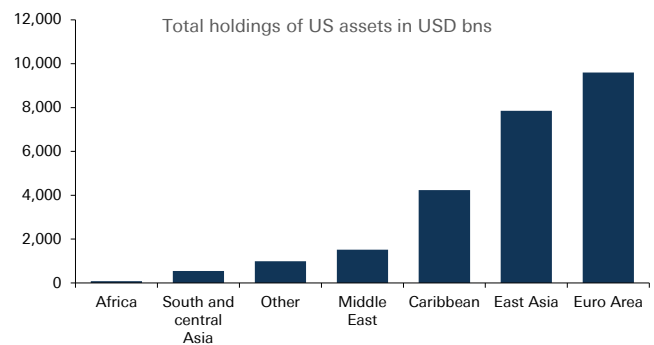
Source : Deutsche Bank, SIPRI Defence Database

Figure 9: Global surpluses driven by similar regions to where defence spending is going to rise the most, global deficits driven by the US



Source : Deutsche Bank, Haver Analytics, *note time series not continuous due to data availability issues

Figure 10: Similarly, holdings of US assets are concentrated in regions that need to re-arm



Source : Deutsche Bank, US Treasury TIC Data

This, however, is logical. With a reduced US security commitment to traditional allies, it is inconsistent for the private savings of those allies to be channeled to the US to the same extent as domestic financing needs increase.

The current account channel: can US defence exports compensate?

Reduced recycling of private savings into the US on account of growing defence spending may not necessarily be negative for the dollar. If US deficits are vulnerable to less recycling of private savings in the rest of the world, the US current account should adjust. But this could happen for dollar-positive reasons if the US is able to provide defence equipment and services through its exports.

The push for defence spending is already evident in the data. For example, although volatile, defence-related machinery orders in Japan and capital goods orders in



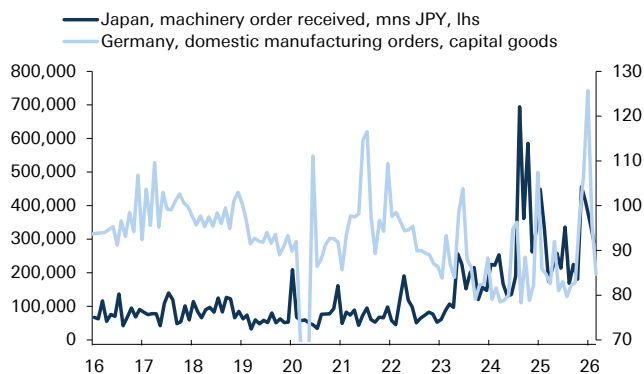
Germany have risen to record high levels in the last four years (figure 11). At the same time, US defence exports have also risen to record levels (figure 12). Nevertheless, while we think that increased US defence exports may compensate the effects on the financing of US deficits in the short term, over the medium term we think this is less likely.

First, although there is a large investment and expertise gap in the defence industry in regions such as Europe, East Asia and the Middle East, which will require imports to fill, over the medium term policymakers are attempting to build up domestic industrial bases and increase strategic autonomy in defence.⁹ For example, the EU has identified seven capability gaps, from air and missile defence to AI, Quantum, Cyber and Electronic Warfare with the aim of building a defence base that will enable the EU to conduct the entire spectrum of military tasks. The European Defence Industrial Strategy has put a target of at least 50% defence procurement being sourced from the European industrial base by 2030 and 60% by 2035. To be sure, the starting point presents a challenge - our European economists estimated that 78% of defence procurement went to non-EU providers of which 63% went to the US at the start of last year.¹⁰ But the European Commission also [note](#) that European defence industry turnover has grown from EUR 124bn in 2021 to EUR 183bn in 2024.

Second, procurement is only one part of the defence budget. For example, although in the EU it makes up the majority of the investment budget, the EUR 88bn spent in 2024 on procurement is less than a quarter of the total (EUR 381bn) spent overall.

Another reason to doubt that US exports will fully compensate for the reduced ability of allies to finance US deficits is that the Trump administration is [itself proposing](#) a major increase in defence expenditure as part of the 2027 Budget, amounting to up to USD 1.5 trillion, nearly a USD 500bn increase from the previous year. It seems unlikely that US industry could simultaneously fully meet global demand for defence equipment and its own domestic demand under government spending plans.

Figure 11: Defence orders in Europe and Japan have already surged in the last three years



Source : Deutsche Bank, Haver Analytics

Figure 12: US defence exports are also climbing rapidly but are unlikely to fully compensate



Source : Deutsche Bank, Haver Analytics

9 See the ReArm Europe Plan, Readiness 2030

10 See Mark Wall and team, "[Focus Europe](#): Defending Europe Part 3, January 2025

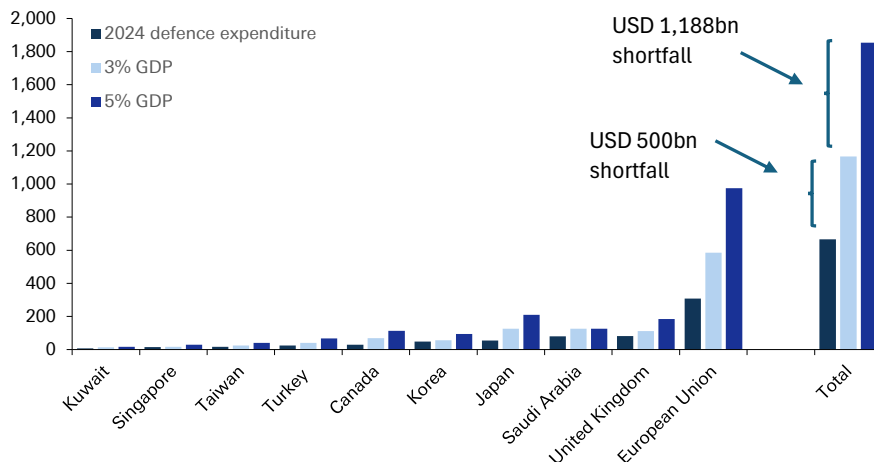


Quantifying the impact

This analysis has suggested that rising defence spending will reduce global savings and compress global imbalances. Because the US deficit is the counter-point to savings in Europe, East Asia and the Middle East, recycling of private savings into the US is likely to diminish, putting pressure on the dollar. In the short term, increased US defence exports may compensate somewhat, but these are unlikely in the medium term to fill the gap.

Can we quantify the impact on USD? Future defence needs are subject to a huge amount of uncertainty but for the sake of simplicity we identify 'defence gaps' among a sub-set of economies where pressure to raise defence spending may be most acute. The first defence gap is based on a hypothetical 3% GDP defence spending target, the second a 5% target. We calculate this based on 2024 GDP in current prices, and compare it to updated defence spending from the SIPRI database for the same year. The results are striking. Under the 3% 'gap', the shortfall amounts to USD 500bn and under the 5% 'gap', the shortfall amounts to USD 1.2 trillion. By way of comparison, the 2024 US current account deficit amounted to USD 1.185 trillion. Of course, this is a highly partial equilibrium analysis, ignoring other aspects of supply and demand for global savings. But at a minimum, it suggests that the effect of increased defence spending on savings rates and current accounts will not be trivial.

Figure 13: Quantifying the defence gap - the additional spending is not trivial



Source : Deutsche Bank, IMF, SIPRI database

Conclusions

This report has focused on providing a framework for how increased global defence spending - a trend likely to be accelerated by the current conflict in the Middle East - will act upon global macro, focusing on savings rates and global imbalances.

We have concentrated on only one sector where investment is likely to rise among US allies over coming years. As noted at the start of this piece, other sectors where investment is likely to rise is energy infrastructure and technology. Our colleagues have focused on other areas in which the furling of the US security umbrella could have major ramifications, for example by reducing incentives to spend dollars in



energy markets or accelerating the transition towards new energy sources.¹¹ When dovetailing with defence spending, the investment in strategic autonomy in these areas will act as another powerful force depressing global imbalances. And even within the defence sector, there are other implications we have not explored, such as the currency invoicing of trade and the multipliers of defence spending. This will provide fruitful ground for further research.

11 Mallika Sachdeva, "[What Iran means for the dollar: a perfect storm for the petrodollar.](#)"



Appendix 1

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Sizing Up GLP-1 Impacts: Survey Shows Durable Shift in Pharma, Diets, and Dining

May 11, 2026

We recently conducted our 2026 survey of current and former GLP-1 weight-loss medication users and the results are clear: GLP-1 adoption is shifting long-term behaviors affecting pharmaceuticals, food and beverage companies, and restaurants.

These themes were already indicated in our initial [2024 survey](#) of GLP-1s, but today's results are even more striking. Now several years into GLP-1 usage, the survey shows sustained changes in dietary and restaurant behavior that stick even after patients have stopped taking GLP-1s. Following the recent introduction of oral pill versions of GLP-1 medications, the survey showed there is strong interest in orals, largely due to the convenience they offer. However, it also shows there are limits related to affordability and long-term commitment.

The takeaway is that, for pharma, there is a plausible pathway for new oral GLP1s to accelerate market penetration over the coming years. Long-run market expansion is likely, although questions remain about how much consumers will be prepared to pay for orals and how long they will stick with them. Meanwhile, regarding food and beverages, consumers are shifting away from large portions and snacking towards protein-forward, functional, and nutrient-dense choices.

For restaurants, the survey points to declines in the number of times people dine outside the home, partially offset by higher spending each time they eat out. Unsurprisingly, brands with differentiated, healthier offerings or more occasion-based demand are likely to be more competitive.

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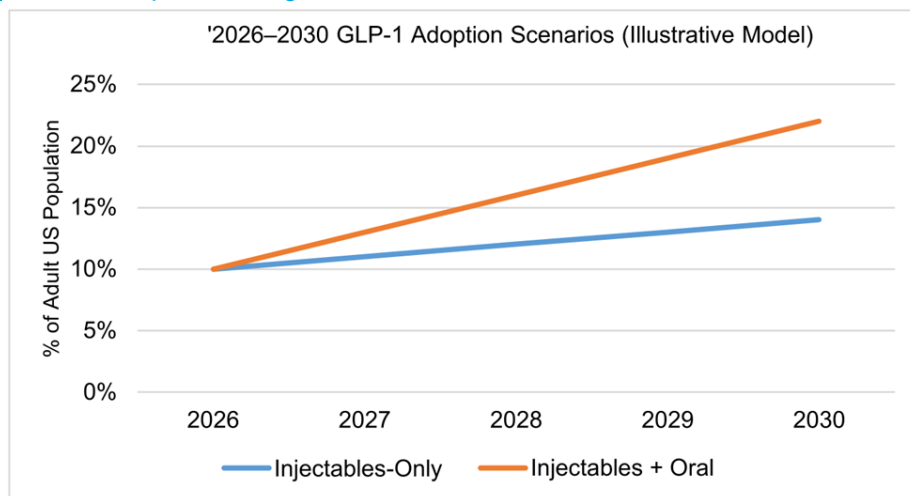
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Figure 1: GLP-1 adoption represents a durable reallocation of consumer demand, in our view; behaviors likely to broaden under accelerated oral GLP-1 adoption through 2030



Source: Deutsche Bank Research

Implications for US and European Pharma

The survey points toward a sustainable promise of oral GLP-1s, but also a cost ceiling, a durability problem, and potential for better risk-benefit to be disruptive. The picture it portrays is of a highly cost-sensitive user base that (still) clearly prefers orals and struggles with adherence to therapy, while mostly intending to return to it at some point and clearly not viewing diet and exercise as a viable alternative.

What that implies for the market outlook is somewhat uncertain: with the up-front caveat that this survey addresses only those trying GLP-1 within the past year and hence perhaps not the key demographic from a market-growth-outlook perspective (i.e., those that have never tried it), for the existing installed user base it implies some risk of injectable cannibalization (though we note this is not particularly apparent yet in prescription data) and also some risk that volume demand may plateau without further price degradation (a minor negative, in our view). However, it also implies that a return to therapy is likely to be high, and improved safety profiles plus oral convenience could help to build out the market opportunity over time. It also suggests the amylin class could help optimize risk-benefit and thus help address the persistent durability problem.

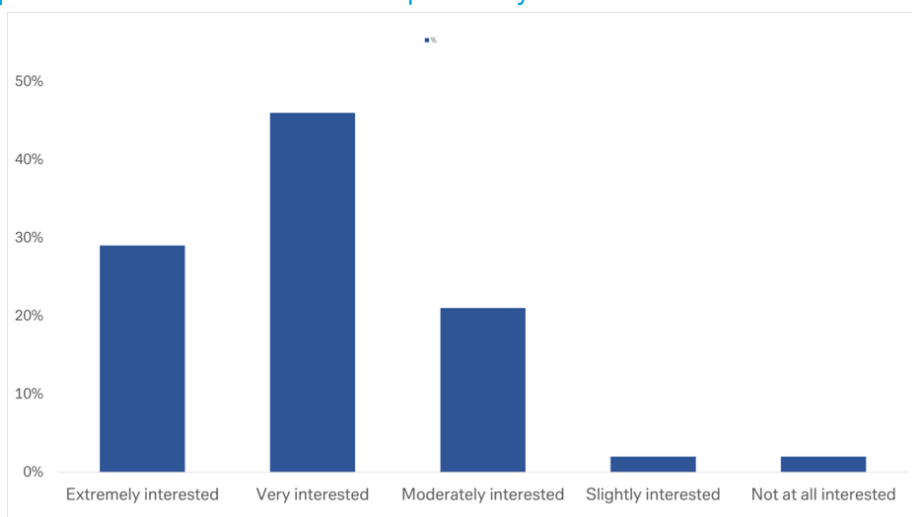
As with our last survey, there was a bias to Ozempic as a preferred brand with a fairly even distribution between Wegovy, Zepbound, and Mounjaro beyond that. Interestingly, the main consequence of increasing use of GLP-1 therapeutics to manage weight loss appears to be a reduction in diet/exercise approaches to helping manage weight loss (rather than the reverse). Beyond that, consistency of adherence appeared static with our last survey (about 70% within a 12-month window), whilst duration of therapy also appeared relatively stable at a mean of 6-9 months. This is despite most patients not being at their target weight loss/implied BMI (current average about 30 versus average target closer to about



25) and many continuing to experience a frequent (20%) rate of GI toxicity of some kind.

Payor coverage did appear to have slightly improved (to about 40%), with most patients still paying up to \$200 per month out of pocket, but interestingly willingness to self-pay has declined to less than half of patients (with most unwilling to pay more than \$200 per month in that scenario). Almost half continue to access therapy via retail pharmacies, with only a modest increase in online channel use. On therapeutic options, interest in oral formulations was very high at about 2/3 (and 80% in younger age groups), with 50% expressing clear preference for orals over injectables all else equal. Interestingly, the greatest sensitivity on the question of switch intent related to cost, with early experience pointing toward a double-digit percentage lower cost for orals versus injectables and most citing this as sufficient to drive switching behavior.

Figure 2: Oral GLP-1 medications are becoming available. With this in mind, please indicate your level of interest in starting to use an oral form should it become an available option for you.



Source: dbDIG Primary Research

The most common reason provided for discontinuation among the approximately 30% who stopped therapy within the past year was cost, though side effects were a close second, and a variety of other reasons followed closely behind. Only about 20% of patients who had discontinued treatment cited "reaching weight loss goal" as the reason. In this subgroup, the advent of oral options also seemed to strike a particular chord (75% extremely or very interested versus 64% in the earlier group of patients still on therapy).

Implications for US and European Consumer Staples

What stood out in our 2026 data compared to our 2024 survey is that GLP-1 adoption is here to stay, with an enduring influence on consumer behavior. The introduction of the oral version of GLP-1s will only reinforce these trends. For packaged food and select beverage categories, the growing prevalence of GLP-1 usage represents a sustained structural pressure that operates through lower



portion sizes, reduced frequency, and selective category substitution rather than outright elimination of consumption.

While many companies have begun to pivot their portfolios in response—introducing protein-fortified snacks, functional beverages, and smaller portion formats—we question whether these innovations can fully offset the net impact of structurally lower intake and frequency. The survey data suggest that many of these shifts improve portfolio relevance and defend share, but do not fully restore lost volume, particularly in indulgent, frequency-dependent categories.

The survey data suggest that protein formats primarily benefit from share reallocation within smaller meals rather than incremental meal creation. Moreover, with traditional categories also chasing protein, we would argue that protein itself is becoming increasingly mainstreamed across everyday food products, diluting the relative advantage of specialized nutrition SKUs and capping category-level growth. Elevated competitive intensity and rising A&P needs increase the risk that theoretical demand alignment does not fully translate into a bigger bottom line.

Outside of performance nutrition, fresh and clean-label portfolios benefit from broader, yet seemingly durable “eat better” behavior rather than calorie avoidance. Survey responses show consistently higher relative consumption of fresh and minimally processed foods during GLP-1 usage, though absolute consumption gains appear incremental rather than transformative.

Flavor-centric and ingredient-led categories appear structurally more resilient than volume-dependent staples. Conversely, sugar-dense, portion-dependent categories face persistent headwinds. Energy drinks, though, stand out as one of the few categories showing net consumption stability or increases among active GLP-1 users. Water and hydration also tend to see benefits, with coffee and tea more neutral.

Overall, GLP-1 adoption increasingly favors goods aligned with satiety, functionality, and quality over calories, but the survey underscores that alignment alone is insufficient. Company-specific execution, competitive position, reinvestment discipline, and cost structure remain decisive in determining whether theoretical portfolio alignment translates into realized financial outcomes.

From a European Food and HPC perspective, our 2026 survey reinforces the view that GLP-1s represent a persistent but comparatively modest influence on consumer demand, rather than a step-change in consumption behavior. In our view, GLP-1 also cannibalizes other forms of weight loss to some degree (though is likely to be more effective and lasting) and as such the net calorie drag per annum that many have worried about seems to be less than feared. It is fair to note though that rolling 12m US Nielsen aggregate “Food” volumes are running down about 1% year-over-year currently and the pace is accelerating. It’s difficult to disaggregate GLP-1 from macro, demographic, SNAP and immigration factors, however, its unlikely to be a non-zero impact.

Importantly, the 2026 cohort appears to have entered GLP-1 usage with healthier baseline habits than in 2023, resulting in a smaller incremental behavioral shift once on medication. In several categories, the change from “before” to “during” GLP-1 usage is less pronounced than previously observed, suggesting an



element of normalization rather than acceleration. That doesn't make the GLP-1 effect itself less valid in its net effect, just that it may not all be related to actual GLP-1 usage. We have also previously noted that the US has also gone through waves of weight loss previously, it's not linear and as such GLP-1 usage increased at a likely time of one of these waves persisting post COVID anyway and we have previously seen a pivot to categories like dairy during non-GLP-1 weight loss.

While some behavioral persistence is evident post-discontinuation, the data do not point to a broad-based collapse in category participation. In fact, select traditionally "indulgent" categories show stabilization—or modest re-engagement—among former users, complicating a simple winners/laggards framework. At an aggregate level, the data are also consistent with lower intensity rather than outright avoidance, implying moderation rather than displacement and an opportunity for companies to tailor their suite of SKUs to mitigate or indeed benefit. As of yet we have seen limited impact on demand for household and personal care categories; thus far, consumer health companies have seen a surprisingly limited number of innovations that support GLP-1 usage.

For our European Beverages coverage we view GLP-1 as a headwind to consumption overall but see some of the survey responses as providing conflicting data points to this potential risk. We view GLP-1 as one of a number of threats to beverage alcohol consumption including 'health', younger consumers, cannabis and, for spirits, smaller players and ready-to-drink. Companies with a higher exposure to the US and those geographies where GLP-1 is coming off patent in the near future (Canada, India, China, Brazil) likely present the greatest risk with beverage alcohol more 'at risk' than soft drinks.

The impact on nicotine consumption is not specifically covered in the survey results. We do recognize that there is growing body of scientific evidence to support the role GLP-1s may play in tackling smoking addiction, but efficacy varies by the specific drug used. The observational data also point to increased effectiveness in reducing nicotine consumption (cigarettes and vapes) although these studies remain relatively small in scale. The industry has long dealt with declining volumes, and so we see GLP-1's as another contributor to the volume contraction but one which, on current evidence, does not represent an existential threat to the industry.

We would also posit the view that the motivation for smoking cessation is typically health related. Increasingly, the industry is able to offer consumers a broad range of smoking alternatives via Next generation products, which significantly improve the health outcomes, compared with smoking.



Figure 3: Categories emerging as relative beneficiaries with the increasing use of weight loss medication

	Category	Description/Details
Relatively Advantaged Categories	Protein shakes and bars	Protein shakes and bars are clear winners. In 2026, 72% of current GLP-1 users consume more than before, up from 64% in 2023, reinforcing protein formats as efficient meal substitutes rather than incremental calories
	Fresh meat and fish	Fresh meat and fish benefit from satiety-led substitution. In 2026, 59% of current users report eating more than pre-medication, up from 45% in 2023, supporting increased prioritization of whole, minimally processed protein sources.
	Fresh produce	Fresh fruits and vegetables remain durable beneficiaries. In 2026, 73% of current GLP-1 users report eating more produce, similar to 2023, indicating sustained "eat better" behavior despite overall moderation in total food intake
	Nuts	Nuts show low downside and solid upside. In 2026, 56% of current users report eating more nuts than before, positioning the category as a functional, nutrient-dense snack aligned with satiety rather than indulgence.
	Condiments, sauces, and spices	Condiments are resilient but not growth-oriented. In 2026, "a lot less" consumption is only 12% while "neither more nor less" rises to 57%, suggesting stability as flavor intensity per bite matters in smaller meals. size.
	Energy drinks	Energy drinks stand out within beverages. In 2026, 47% of current users report drinking more than before, while only 33% report drinking less, supporting functional, zero-sugar positioning over indulgent beverage formats.

Source: Deutsche Bank Research

Figure 4: Categories impaired by GLP-1 usage but more resilient, especially with adaptation to remain competitive

	Category	Description/Details
Mixed / Execution-Dependent Categories	Packaged and frozen meals	Frozen and refrigerated meals sit at the execution frontier of GLP-1 impact: neutrality remains the modal outcome, with incremental—not structural—pressure, hinging on portion control, protein density, and clearer health cues
	Canned foods and traditional pantry staples	Pantry staples are neutral-to-slightly pressured, not broken: consumption skews toward "same or less," with limited upside absent reformulation, protein enrichment, freshness cues, and improved portion transparency
	Coffee	Coffee is compositionally resilient rather than volume-challenged: core consumption holds steady, while sugar- and syrup-heavy formats decline—shifting pressure to sweetened customization, not coffee occasions themselves
	Tea	Tea is one of the most resilient categories in the dataset: high neutrality, minimal cutbacks, and muted intent-to-reduce reinforce its role as a low-calorie, habitual utility beverage insulated from GLP-1 moderation dynamics

Source: Deutsche Bank Research



Figure 5: Categories that are more adversely impacted by the use of weight loss medication

	Category	Description/Details
More At-Risk Categories	Chocolate and candy	Chocolate and candy face the clearest structural pressure. Roughly half of current GLP-1 users consume less in 2026, and most former users fail to rebound, confirming sustained demand erosion in calorie-dense indulgence
	Cookies and biscuits	Cookies and biscuits remain highly exposed. One-third of current users report eating "a lot less" in 2026, while ~75% of former users continue to eat less or stop entirely, reflecting vulnerability of habitual, low-satiety snacks
	Ice cream	Ice cream participation is persistently impaired. About 70% of current users report same or lower consumption, and a similar share of former users continue to eat less or none, signaling weak post-GLP-1 recovery
	Salty snacks	Traditional salty snacks face uneven but meaningful pressure. Chips consumption declines, with "a lot less" rising to 17% in 2026; pretzels show limited rebound, reinforcing structural headwinds for refined-grain snack formats
	Cereals and refined pantry staples	Cereals show elevated "eat less" signals during GLP-1 usage and weak normalization afterward. Their association with refined carbohydrates and low satiety leaves cereals structurally disadvantaged versus protein-dense alternatives
	Alcohol (Beer, Wine, Spirits and RTD cocktails)	Alcohol faces moderate but notable headwinds. Participation declines materially (spirits, beer, wine), yet "drink more" exceeds "drink less" during treatment, suggesting reduced frequency rather than category exit and a nuanced, uneven risk profile

Source: Deutsche Bank Research

Implications for Restaurants

The increasing penetration of GLP-1 weight loss drugs is intuitively a headwind for total restaurant spend, and our survey results are in line with this thinking. Despite the fact that many users eventually stop taking GLP-1s, they continue to show sustained changes in behaviors. Our survey results indicate that former GLP-1 users do not meaningfully return to full-service restaurants after taking the medication, though some return to fast food and coffee shops at a rate above current GLP-1 users, and in reality, we believe restaurant trends rebound more than our survey suggests. This is based on anecdotal commentary and behaviors seen in other categories. The read-through on individual frequency was more positive, with current GLP-1 users reporting an increase in visits to full-service restaurants, and a modest decline in visits to fast food, coffee shops, and takeout relative to pre-GLP-1 levels. Moreover, average spending on meals per visit was higher than baseline, suggesting a more tempered impact on sales compared to traffic.

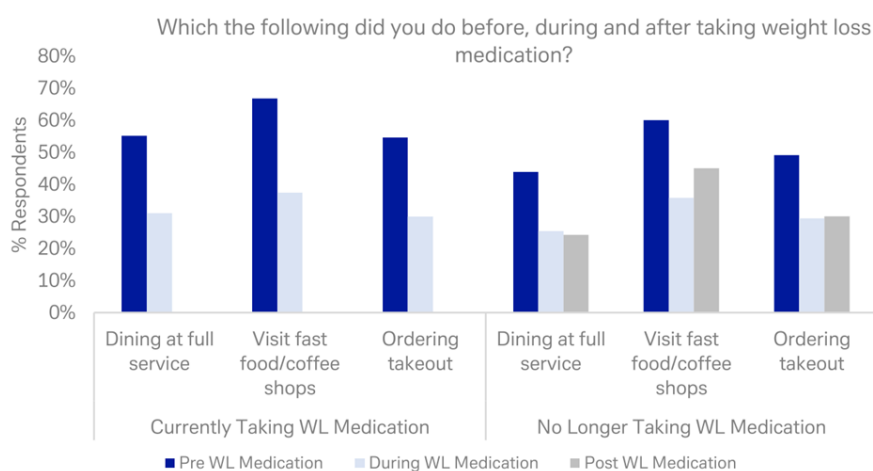
Restaurants have started to address behavioral shifts by offering smaller portion sizes, protein-forward options and functional beverages. We believe these innovations can help more effectively sustain participation in the restaurant industry; brands that are differentiated, screen healthy, and are more occasion based are relatively best positioned to be competitive.

It is clear that overall food consumption decreases when consumers are on GLP-1 medications, though the implications once consumers transition off are less clear. Based on our survey, 35% consumers report eating more on GLP-1 medications, 24% eat about the same, and 41% report eating less, for a net 6% eating less. When consumers were asked about current eating habits relative to pre-GLP-1 medication, among those currently taking GLP-1s, a net 10% reported eating less compared to pre-medication (38% more, 48% less). Notably, among consumers that are no longer taking GLP-1s, when compared to prior behaviors, a net 20% indicated they are actually eating more (45% more, 24% less), suggesting former users revert toward prior aggregate consumption levels.



That said, there is still a change in how consumers eat, most notably an increase in cooking. Based on our survey, among consumers currently taking GLP-1s, a net 54% of consumers are cooking more at home. Cooking usage also remains elevated among consumers no longer taking GLP-1s, with a net 47% of consumers cooking more at home than before medication.

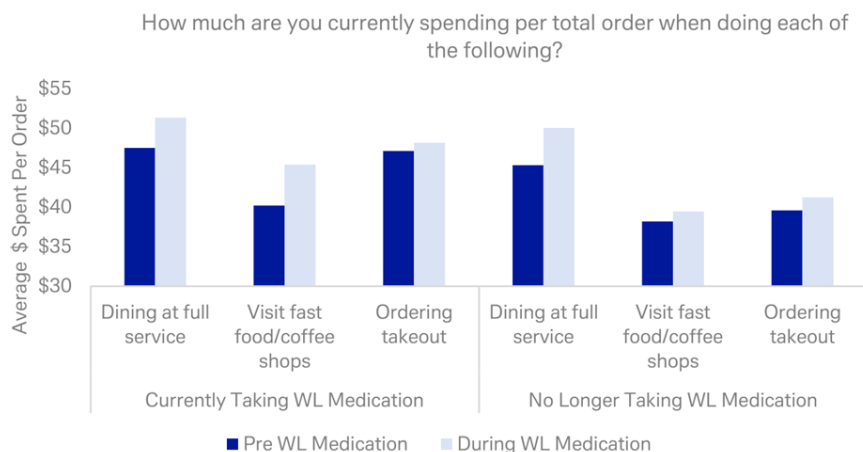
Figure 6: Our survey shows a meaningful and broad-based decline in restaurant usage when consumers take GLP-1s. Among current GLP-1 users, the percentage of respondents that visit restaurants drops by 25-30pp versus pre-weight loss behavior. Among former GLP-1 users, the percentage of respondents that visited restaurants while taking these medications versus prior behavior decreases at a relatively similar 20-25pp.



Source: dbDIG, Deutsche Bank Research

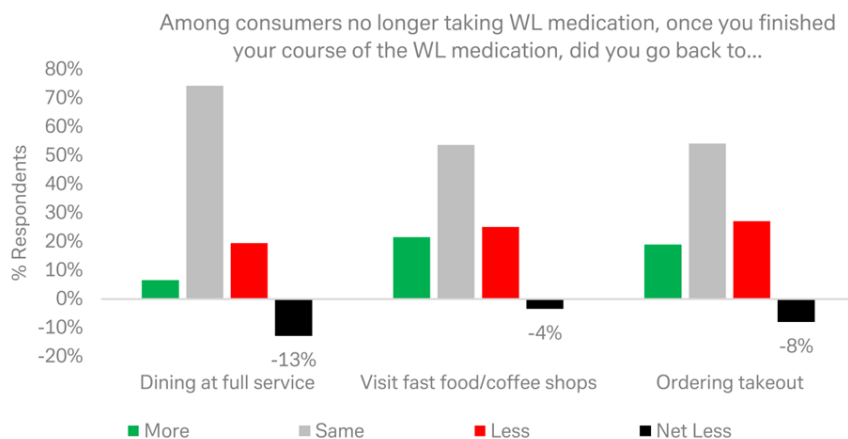


Figure 7: Among GLP-1 users that regularly visit restaurants, average spend per month did not take much of a hit, noting both current and former GLP-1 users report higher average spend per visit for full service, fast food/coffee shops and takeout. It is unclear whether the higher average spend per visit is a reflection of perceptions around higher prices or consumers ordering more/trading up.



Source: dbDIG, Deutsche Bank Research

Figure 8: Individual consumption for former GLP-1 users that are still eating at restaurants seems reasonably resilient (or at least manageable). On a net basis, 13% of consumers said they were dining less at full service compared to pre-GLP-1 behaviors, 4% said they were visiting fast food/coffee shops less, and 8% said they were ordering takeout less.



Source: dbDIG, Deutsche Bank Research



The Bottom Line: A Lot to Digest

Our survey implies that GLP-1s are starting to look less like a therapeutics cycle and more like an enabling platform. It is reshaping how consumers manage weight, what they eat and drink, and whether and how they choose to dine out. For pharma, orals offer an expanded market, offset to an extent by affordability and other constraints. For consumer staples and restaurants, brands that can focus on health-forward innovations will remain more competitive in this new environment.

Research Methodology

This study focuses on both current and past GLP-1 users, with demographic data collected via a dedicated set of questions in our US monthly survey. Compared to the 2023 wave, several demographic shifts were observed:

Gender: The user base is becoming more gender-balanced, with male GLP-1 users increasing from 40% to 45%.

Income: While the overall income distribution remained stable, there was a marginal increase in GLP-1 users within Income Tier 1 (households with an income up to \$25,000).

Overall, the survey was conducted through dbDIG from March 20-30, 2026, with 550 respondents. We would like to thank Anthony Chaimowitz for his significant contribution to the survey and this report.



Appendix 1

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May 5, 2026

Q&A with Matthew Barnard

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As equity research adapts to rapid shifts in technology, markets and client behaviour, Matt Barnard, Head of US Equity Research, is helping steer how Deutsche Bank evolves its research offering. In this edition of our Q & A series, Matt shares his perspective on the new ways clients are consuming research, his outlook for US markets, and what it's like leading research teams through a period of significant change.

Q: You've spoken about the equity research landscape undergoing a fundamental shift. What has changed most in how clients discover, engage with and use research today?

AI is fundamentally changing how clients consume sell-side research. As information becomes more easily accessible through AI tools, the value of differentiated insights from trusted sources has moved higher up the value chain. Investors are increasingly using AI to discover new ideas and challenge consensus views more quickly and efficiently. This allows them to filter out noise and spend more time engaging in deeper, more meaningful conversations with our Deutsche Bank research analysts.

Q: Against a complex backdrop for US geopolitics and policy, how do you currently assess the outlook for US markets – and which themes matter most for clients as we move through 2026?

Geopolitical developments in the US and globally continue to be a significant driver of volatility across risk assets. Deutsche Bank research's macro and strategy teams have consistently taken bold, often counter-consensus views, which are especially valuable in this environment. While conflicts in the Middle East will likely continue to create ripple effects across markets, the underlying fundamentals for US equities remain relatively strong. The rapid acceleration of AI adoption – and the associated capital investment – remains a key thematic focus for clients, both in the US and globally.

Q: You now lead teams through a period of rapid change. What do you find most rewarding about working in research management at this point in the cycle?

At its core, the role of an equity research analyst is to form forward-looking views with imperfect information. The strongest analysts thrive in periods of uncertainty and rapid change, identifying emerging trends and uncovering new investment opportunities early. From a management perspective, the



Q&A with Matthew Barnard

most rewarding part of my role is seeing some of the brightest minds in research stay ahead of these shifts and deliver true alpha for our clients.



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The long road: Quantifying Europe's exposure to transport disruption

May 8, 2026

Heightened geopolitical risk in the Middle East has placed European transport networks under intense scrutiny, with a particular focus on the supply of jet fuel given the high share of the Gulf in the global supply of kerosene.

For the supply side of the European economy and overall input availability, the risk from a shortage of jet fuel is relatively small. The dominant fuel dependency for goods production is on diesel and electricity, reflecting the four-fifths share of input costs taken by land transport (road and rail).

The true vulnerability to a shortage of jet fuel lies squarely with the tourism sector. Our simulation shows a 10% drop in tourism flows would have a moderate impact on the Euro Area, on average (-0.2% of GDP), but a more significant impact on countries like Croatia (-1.5%), Greece, and Portugal (-0.7%).

Beyond the specific risk to jet fuel, a wider disruption to the global supply of transport would expose a more fundamental European vulnerability: the physical length of supply chains. To measure this vulnerability, we introduce a “physical distance travelled” metric, reflecting how far inputs travel for every industry. It reveals two critical points:

- A clear geographic divide: The economies with the longest supply chains are generally located in Southern Europe. Lacking deep integration with Europe's industrial core their value chains are inherently more global and reliant on long-distance sourcing. This exposure to transport frictions could create a vulnerability for some of the zone's high-debt member states. These countries are also those we previously found to have the highest energy import bill sensitivities.
- A dual "hidden" risk: From a sector perspective, the most-exposed sectors besides tourism are capital goods (electronics, air/spacecraft, machinery) and materials manufacturing (metals, refineries), whose supply chains stretch nearly 5,000 km, on average. Crucially, these are the same value chains we [previously identified](#) as having high "hidden" energy content, indicating a dual vulnerability to both transport shocks and embedded energy costs.

What does this mean for the ECB? Our baseline view is a 'measured' tightening with 50bp of hikes to 2.50%, the upper end of the range of neutral. With the labour market cooling, financial conditions tightening and this threat of fuel shortages, the growth outlook is not robust enough to think the ECB hikes much beyond 2.50%. The challenge for the ECB could be fragmentation if fuel shortages have a differential impact on member states.

Authors

Yacine Rouimi
Senior Economist

Mark Wall
Chief Economist



Concerns shifting from energy to transport. Aviation of particular concern. As we highlighted in a recent [note](#), Europe's economic exposure to the Gulf conflict is twofold, and is more complex than the singular energy shock of the Russia-Ukraine war. Risks of direct shortages in energy and key materials like petrochemicals or mineral fuels are now coupled with a more complex threat: second-order disruptions to global value chains.

Transport disruption is one of the most significant vectors of transmission for both these risks. As an input at every stage of production, the transport sector's resilience is once again at the forefront amid reports of looming [fuel shortages and logistical bottlenecks](#). In all this, jet fuel is of particular concern, as shown by recent statements by the Head of the International Energy Agency on April 16 that Europe has "[maybe six weeks of jet fuel left](#)", and our own [calculations](#) that the share of Gulf countries as a source for EU imports of kerosene reaches 56%.

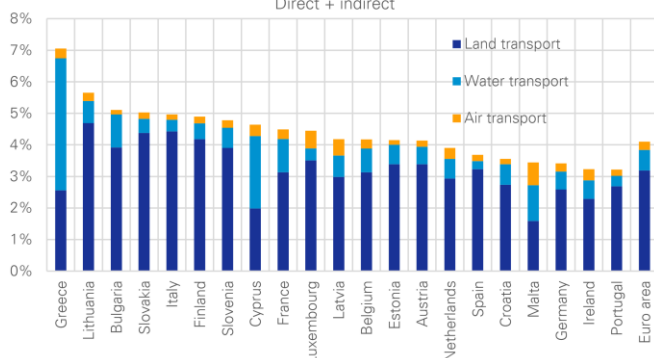
As our transportation sector specialists explain, the market is already under pressure. Jet fuel prices have doubled, and while replacement barrels from the US and Africa have helped Europe plug the gap for now, this fragile balance is unlikely to survive the summer demand surge. Airlines are already cancelling unprofitable flights and face difficult trade-offs between raising prices and eroding consumer confidence.

In this article, we think about the European economy's exposure to potential global transport disruption – in particular jet fuel – from two complementary perspectives. First, from a supply or production perspective, we look at the role of various forms of transport in European supply chains, including air transport, while introducing a distance metric reflecting how far European inputs travel and are exposed to transport disruptions. Second, from a demand or income perspective, we look at the exposure of European economies to tourism and in turn tourism's exposure to air travel.

How much transport in value chains? 3-5% of inputs, of which little air transport.
A traditional way to assess transport dependency is to measure its overall share in inputs along the value chain.

Figure 1: Cost share of transport services in Euro area input purchases, dominated by land. Air is tiny

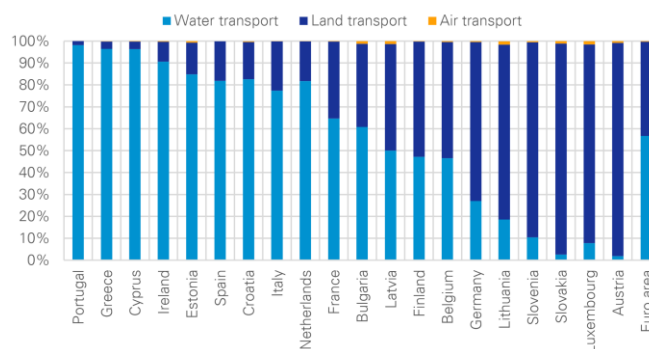
Euro area countries - Share of transport services in overall input costs - Direct + indirect



Source: Deutsche Bank Research, OECD ICIO. The figures include the absorption of all transport services at every stage of production, including that required to move inputs between two foreign suppliers (ie indirect absorption).

Figure 2: By physical weight, maritime shipping dominates freight transport

Euro area - Freight transport volume - By mode of transport



Source: Deutsche Bank Research, Eurostat (tran_hv_ms_frmod). The unit used for measuring the freight transport shares is the tonne-kilometre (tkm). One tonne-kilometre represents the movement of one tonne over a distance of one kilometre.



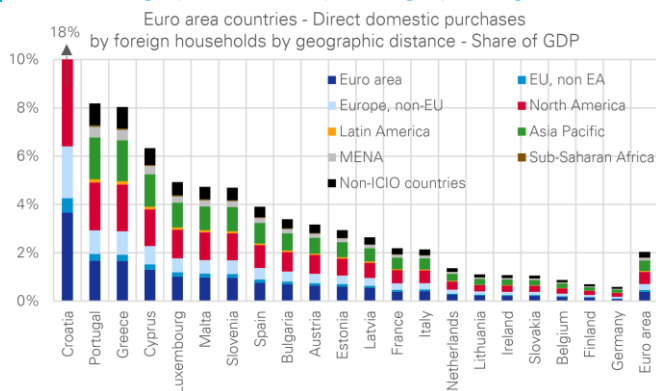
Our calculations, using a methodology [popularized by the OECD](#), suggest transport services account for 3-5% of total input purchases across the Euro area (Figure 1), with some notable outliers like Greece at 7%. Among the Big4, the role of transport is more pronounced in Italy and France (4.5-5%), less so in Germany and Spain (3-4%).

Land transport constitutes the bulk of this transport cost – approximately four-fifths. The share of water transport is significantly smaller, except for trade-reliant coastal nations like Greece, Malta, and Cyprus. Importantly, air transport represents only a minor share overall.

Implications for fuel usage. The dominance of land transport (i.e., road and rail) in the cost structure implies that the most significant fuel availability risk in the supply of goods and services comes from disruptions to diesel and electricity, not jet fuel. Given air freight's minimal share in both costs and physical volume of freight (as confirmed by both our analysis and Eurostat data in Figure 2), even a severe, un-substitutable 10% shortage of jet fuel may not cause much disruption to the flow of inputs and economic production in Europe.

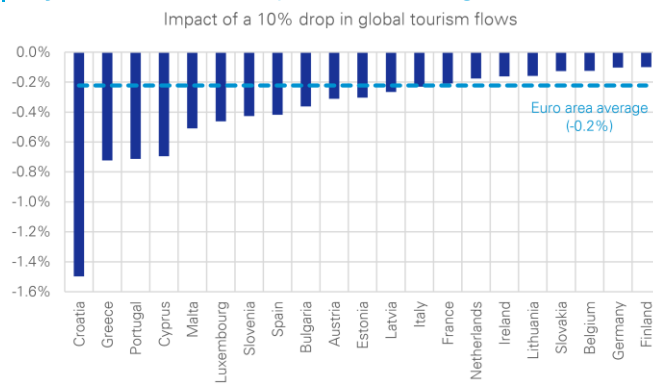
Tourism: a distinct, service-based vulnerability. The above numbers, focusing on freight and input purchases, do not capture the broader issue of the tourism economy: tourism involves moving individuals – not freight – and air transport is not counted as a formal input to the hotels, restaurants, and cultural activities that constitute the tourism economy (i.e., hospitality, recreation), but generally as a purchase by households.

Figure 3: The high share of long-distance travellers in direct foreign purchases (spending by foreign tourists)



Source: Deutsche Bank Research; OECD ICIO.

Figure 4: Tourism is where the Euro area's vulnerability to jet fuel and air transport becomes significant



Source: Deutsche Bank Research, OECD ICIO. The simulation focusses on the impact of spending by foreign tourists, and does not capture domestic tourism. As such, the numbers differ from the estimates of the sector's footprint published by Eurostat

In Figure 3, we calculate the share of GDP driven by direct purchases by foreigners (spending by foreign tourists). The result, unsurprisingly, is that the nations of Southern Europe are much more exposed, on average, but all nations in Europe derive income from a demand perspective from far-away visitors. We estimate that about three quarters of that spending is made by foreigners having travelled more than 2,000 km (50% of the spending by travellers more than 5,000 km away, often placing them on another continent).



A simulated 10% drop in global tourism flows reveals a manageable cost for the Euro Area, on average (a 0.2% hit to GDP – Figure 4). The impact is more pronounced in Croatia (GDP falling 1.5%), and Greece and Portugal (both seeing declines of 0.7%). On the Big4 countries, Spain would be faced with the most significant loss (0.4%), while France, Italy (0.2%) and Germany's (0.1%) exposures are more modest.

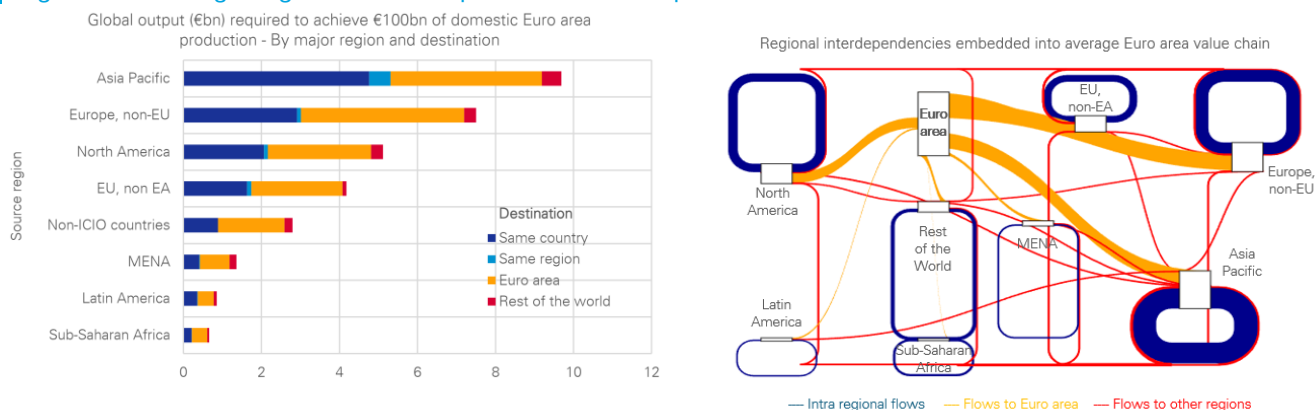
An important caveat to bear in mind: in cases of severe disruption of air travel, holidaymakers would likely travel more locally, which should reduce the size of a shock to the tourism industry. The overall impact will still depend on the size of the tourism sector relative to local demand.

Beyond jet fuel: Measuring exposures to transport friction. The focus on jet fuel, while critical for tourism, highlights a greater strategic risk: energy-driven transport disruptions may not stop at kerosene. A broader shock to transport fuels affecting diesel and maritime fuels would more certainly curtail the physical movement of goods. In such a scenario, what matters is not just the cost of transport, but a value chain's fundamental exposure to distance.

To assess this exposure, we introduce the “physical distance travelled” metric. This measure goes beyond a simple 'average import distance' to quantify the total physical journey embedded in the Euro Area's inputs to production, including domestic trade and all transactions that occur before a final good is imported. It reveals the true length – and vulnerability – of our supply chains. Deutsche Bank clients can access further details including the methodology on our [Deutsche Bank Research](#) client site.

Figure 5 illustrates the main input to the calculations. It shows the global transactions required to generate €100bn of production of an average good/service in the Euro Area, and where this output is directed. Both charts contain the same data, with the left-hand one showing how output is broken down (domestic consumption vs. exports), while the right-hand one illustrates the complex connections across regions to feed the Euro Area with inputs. The blue and red links represent the Euro Area's indirect exposure.

Figure 5: Picturing the global web of inputs for Euro area production



Source: Deutsche Bank Research, OECD ICIO. Note: our distance metric also includes the intra Euro area trade required in the value chain, not displayed in this chart.



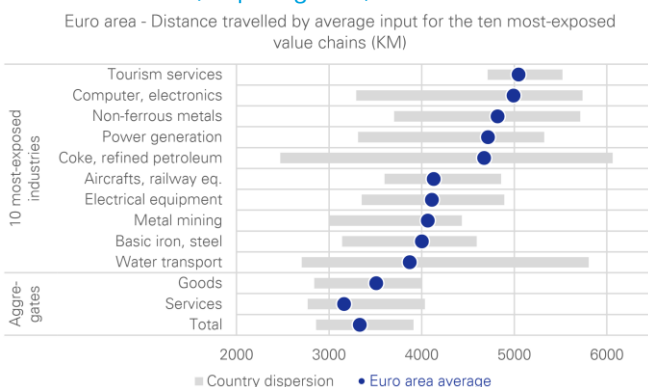
We find a significant reliance on external partners, requiring approximately €10bn in output from the Asia-Pacific region alone. The charts also highlight the complexity of these value chains: a large portion of the Asian output is transacted and transformed internally (the blue segments) before being exported.

Tourism is clearly at risk, while the capital goods and materials value chains stand out on the goods-producing side. Southern Europe is more vulnerable, too.

The distance metric we calculate is simply the sum of the distances between the nodes in Figure 5 in kilometres, weighted by their values (it also includes intra euro area trade, not displayed in the chart). It shows that the longest, and, therefore, most transport-exposed, value chains in the Euro Area fall into three clear categories:

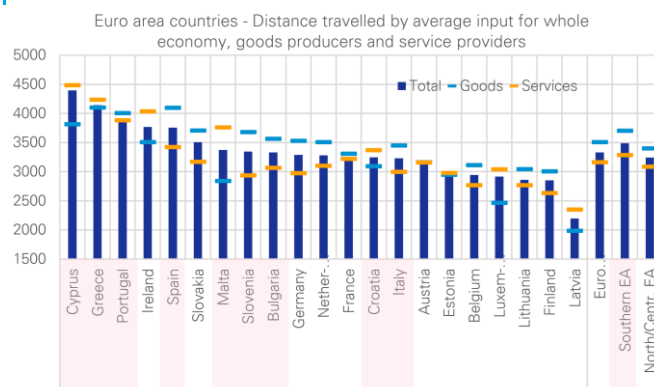
- First are tourism services, with most countries deriving a significant portion of revenue from travellers residing more than 5,000 km away (Figure 6).
- Second, Capital goods, where production is globally fragmented. Here, the computer and electronics sector stands out, with the average input traveling nearly 5,000 km, the longest of any European industry, after tourism. Other capital goods sectors like aircraft/railway equipment and electrical equipment (4,100 km) also exhibit noticeably long supply chains.
- The third category is materials manufacturing. Industries like non-ferrous metals (4,800 km), coke/refined petroleum (4,700 km), and iron/steel (4,000 km) all rely on sourcing raw materials from distant global locations. Notably, these are the same value chains we previously identified as having a high ["hidden energy"](#) imported content, indicating a dual vulnerability to both energy shocks and transport disruption.

Figure 6: Highest distance dependency for European tourism services, capital goods, and materials.



Source: Deutsche Bank Research, OECD ICIO

Figure 7: Longer distances in Southern European value chains



Source: Deutsche Bank Research, ICIO

In terms of geographies, the simulations show that several Southern European countries have longer-than-average supply chains: a group including Cyprus, Greece, Portugal, and Spain are all positioned at the higher end of the spectrum (Figure 7). An average calculated for the nine Southern Euro area economies

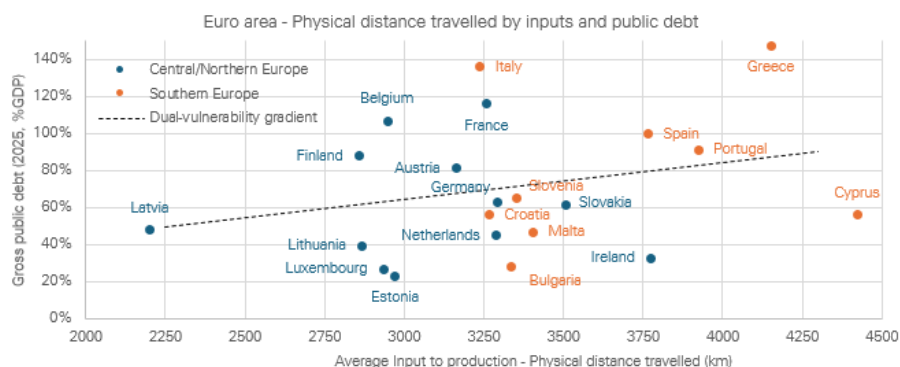


yields a kilometric distance of 3,490 km, a figure higher than the Euro Area average of 3,330 km. For goods producers within this southern bloc, the average distance is even longer, at 3,700 km.

The higher exposure to transport frictions could, therefore, create a vulnerability for some of the zone's high-debt member states (Figure 8). These countries are also those we previously found to have the highest energy import bill sensitivities.

Italy and Croatia are exceptions to this pattern, showing shorter supply chain distances more aligned with the industrial core of the Euro area, like Germany and France (3,200-3,300 km).

Figure 8: A vulnerability to transport disruptions for some of the highest-debt member States



Source: Deutsche Bank Research, OECD ICIO

What this means for the ECB: Measured tightening...at most

A negative supply shock pushes prices up and activity down. These opposing forces can neutralise the impact on monetary policy ('look through'). The ECB is nervous that the size and persistence of the upside inflation shock might need a tightening of monetary policy to ensure that inflation remains anchored at 2% in the medium term. However, the downside risks to growth are not irrelevant. The more significant the downside risks to growth, the less likely is a large or persistent inflation shock. Monetary policy needs to carefully weigh the inflation effects against the growth effects.

If Europe suffers from an outright shortage of transport fuels, then growth will undoubtedly weaken more than expected (assuming no substitutability of fuels or other sources of transport). Shortages could also push transport prices up even more, but only if firms believe demand can sustain even sharper price hikes, and this cannot be taken for granted – especially when the fiscal response to the crisis is limited and financial conditions are tightening.

In other words, if fuel supply shocks intensify to the point of outright shortages in Europe, it is not obvious that monetary policy would have to tighten even more than expected. It is even possible that monetary policy tightens less.



Our view is that the downside risks to growth are becoming more significant. One reason is the impact of the energy shock on supply chains. As discussed above, there are multiple aspects to this shock on supply chains.

First, we recently wrote about the hidden energy costs in Europe's imports, particular from Asia, a region where the risk of production disruptions from energy shortages is more acute.

Second, last year, we wrote about the high complexity (and low substitutability) of Europe's imports from Asia (China in particular). This increases the risk of an Asian shock compromising European supply chains.

Third, in this note, we offer a complementary perspective by showing the actual length ('physical distance travelled') of Europe's supply chains. The member states most at risk of an economic cost from a limited supply of transport fuels are those with longer supply chains. Generally speaking, they are also the ones with higher levels of public debt.

The bottom line is we see the net inflation/growth arguments favouring a measured ECB tightening at most. Our baseline is 50bp of hikes to 2.50%, with quarter point hikes in June and September. With the labour market cooling, financial conditions tightening and a threat of fuel shortages, the growth outlook is not robust enough to think the ECB hikes much beyond 2.50%. The challenge for the ECB could be fragmentation if fuel shortages have a differential impact on member states.



Appendix 1

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